



Systemic risks of dominant online platforms: A scoping review

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ABSTRACT

The 2022 EU Digital Services Act (DSA) is a major milestone in the regulation of online platforms and search engines, and the way that they affect democratic societies. At its core, the DSA lays down rules and obligations for online platforms and search engines to identify, assess and mitigate “systemic risk” arising from their design, functioning or use. Although it is the DSA that introduces this term in EU platform regulation, its definition has been criticised as being overly broad and providing limited legal clarity. In response, we conducted a PRISMA-based scoping review and thematic analysis of past literature and case law concerned with the definition of “systemic risk.” This literature review reveals key themes inherent to systemic risks to clarify when a risk crosses the threshold of systemic: structural characteristics, sources, and societal impact of systemic risks. Upon this basis we develop a framework to guide researchers, policymakers and platforms in assessing when a risk triggers the “systemic” threshold. This article thereby contributes to the literature by providing a sound theoretical foundation for systemic risks to fundamental rights arising from online platforms.

1. Introduction

We have increasingly become reliant on various intermediary services to navigate our lives: From search engines to social media and e-commerce platforms, intermediary service providers act as a gateway to the Web. In this way, providers such as Google, Meta, and Amazon influence both how we produce and consume, raising crucial considerations around the power to regulate and steer online and offline behaviours.¹ Because of this, European policymakers consider that these intermediaries bear responsibility for ensuring a “safe, predictable, and trustworthy online environment” and have codified their responsibilities with the Digital Services Act (DSA).²

The DSA introduces two distinct categories of intermediaries: Very

Large Online Platforms (VLOPs) and Very Large Online Search Engines (VLOSEs). VLOPs and VLOSEs are online platforms and search engines with an average monthly active user base in the Union equal to, or higher than, 45 million.³ This designation reflects the significant influence that these intermediaries hold online.⁴ As a rule, intermediary service providers must adhere to duties and obligations that are proportionate to the specific risks that may arise from their services.⁵ In other words, the DSA recognises the link that exists between the scale of a service and its potential to cause harm: VLOPs and VLOSEs must meet stringent additional obligations to ensure the safety of the digital spaces they govern, and provide adequate safeguards for fundamental rights.⁶ These additional obligations include the regular assessment of systemic risks and the implementation of measures to mitigate those risks.⁷ This

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¹ Elettra Biatti, “A Genealogy of Digital Platform Regulation” (2023) 7 Georgetown Law Technology Review 1, 25; Julie E Cohen, “Law for the Platform Economy” (2017) 51 UC Davis Law Review 136–37.

² Recital 3 Regulation (EU) 2022/2065 of the European Parliament and of the Council of 19 October 2022 on a Single Market For Digital Services and amending Directive 2000/31/EC [2022] OJ L 277/1 (DSA) 2022.

³ Article 33(1) *ibid*.

⁴ Mark Leiser, “Reimagining Digital Governance: The EU’s Digital Service Act and the Fight Against Disinformation” (2023) 4 <<https://doi.org/10.2139/ssrn.4427493>> accessed June 1, 2025.

⁵ Giovanni De Gregorio and Pietro Dunn, “The European Risk-Based Approach: Connecting Constitutional Dots in the Digital Age” (2022) 59 Common Market Law Review 313, 484.

⁶ see Recital 75 DSA; Sally Broughton Micova, “What Is the Harm in Size?” (Centre on Regulation in Europe 2021) 5; Leiser (n 4) 4.

⁷ Articles 34, 35 DSA.

risk governance model thus aims to address the complex risks and harms associated with the services and structure offered by dominant intermediaries.⁸

Risk management obligations have been recognised as the cornerstone of the DSA rulebook for VLOPs and VLOSEs.⁹ According to Article 34(1) DSA, VLOPs and VLOSEs must “diligently identify, analyse and assess any systemic risks in the Union” which arise from the “design or functioning of their service and [...] systems.” The risk assessment conducted must be “specific to their services and proportionate to the systemic risks, taking into consideration their severity and probability,” with systemic risks including the dissemination of illegal content, any actual or foreseeable negative effects for the exercise of fundamental rights, for civic discourse and electoral processes, or in relation to gender-based violence, the protection of public health and minors and serious negative consequences to the person’s physical and mental well-being. Article 34(2) DSA specifies that factors to consider in the risk assessment include the design of its content moderation or recommender system, data related practices, or their terms and conditions.

Despite the central role of “systemic risks” in the DSA, the concept has been criticised for being too “open-ended”.¹⁰ Leerssen argues that while the DSA offers a long list of factors to aid the assessment of systemic risks and possible mitigation measures, it fails to set out the actual substantive norms following from risk management obligations. By describing four possible types of systemic risks, the DSA follows a “contextual” application of systemic risks and does not provide a comprehensive definition of the notion.¹¹ This gap was also not addressed in the later 2024 Guidelines on the mitigation of systemic risks for electoral processes.¹²

Given the central role of systemic risk for the DSA, a clear definition is essential to establish when its risk mitigation tasks are triggered.¹³ At the same time, this lack of clarity prompts fundamental rights concerns,¹⁴ in a context where fundamental rights are the “constitutional foundation” of European integration.¹⁵ The lack of specifications regarding how areas of tension between fundamental rights (e.g., protection from harmful speech vs. freedom of speech) should be reconciled

represents an obstacle in developing the “safe, predictable, and trustworthy online environment” set out by the DSA.¹⁶ As the wording makes it difficult to establish how systemic risks will be interpreted, the actual impact that the DSA will have on fundamental rights remains unclear.¹⁷ Yet, despite calls from scholars¹⁸ and civil society¹⁹ to define systemic risks, there is still no actionable framework for identifying these risks under the DSA that is grounded in a thorough review of how systemic risks affect fundamental rights.

In this article, we fill this research gap by investigating how the concept “systemic risk” has been defined in past literature and case law. We accomplish this by systematically carrying out a scoping review. The aim of this review is not to systematise the general risks that have been identified in the literature as arising from platforms and online services. Instead, we seek to supplement this scholarship by examining when and how the “systemic” threshold is triggered. Additionally, we introduce a framework that offers practical support for researchers, policymakers and platforms to determine whether a risk has become systemic for the purposes of the DSA. The research question guiding this paper therefore is: *When does a risk become systemic, according to past academic literature?*

In the following, we present, in Section 2, the existing background research that motivates us to conduct a scoping review of systemic risks. In Section 3, we elaborate on our methodological approach. In Section 4, we present the results of our review and discuss, in Section 5, our framework and the implications of our research findings. We focus both on the state of the literature before the DSA became relevant, and on the state of the literature since the DSA proposal until November 2025.

2. Background

While the concept of systemic risk has commonly been used in other sectors (e.g., financial markets regulation, healthcare, or environmental literature) the inclusion of this term in platform regulation is novel.²⁰ This shift reflects the growing acknowledgment that the rapid digitisation of nearly every aspect of modern life brings significant risks, leading to a more prominent role for the principle of risk in regulating digital markets and society.²¹ In the following we draw upon one of these

⁸ Ilaria Buri and Joris van Hoboken, “The Digital Services Act (DSA) Proposal: A Critical Overview” (2021) 32 <https://dsa-observatory.eu/wp-content/uploads/2021/11/Buri-Van-Hoboken-DSA-discussion-paper-Version-28_10_21.pdf> accessed June 1, 2025; De Gregorio and Dunn (n 5) 497.

⁹ Paddy Leerssen, “Counting the Days: What to Expect from Risk Assessments and Audits under the DSA – and When?” (DSA Observatory, January 30, 2023) <<https://dsa-observatory.eu/2023/01/30/counting-the-days-what-to-expect-from-risk-assessments-and-audits-under-the-dsa-and-when/>> accessed June 1, 2025.

¹⁰ *ibid.*

¹¹ Zohar Efroni, “The Digital Services Act: Risk-Based Regulation of Online Platforms” [2021] Internet Policy Review <<https://policyreview.info/articles/news/digital-services-act-risk-based-regulation-online-platforms/1606>> accessed June 1, 2025.

¹² Commission Guidelines for providers of Very Large Online Platforms and Very Large Online Search Engines on the mitigation of systemic risks for electoral processes pursuant to Article 35(3) of Regulation (EU) 2022/2065 [2024] C/2024/3014.

¹³ Alexander Peukert, “Response to the Public Consultation: Guidelines for Providers of Very Large Online Platforms and Very Large Online Search Engines on the Mitigation of Systemic Risks for Electoral Processes” (Goethe Universität Frankfurt 2024) 3 <https://www.jura.uni-frankfurt.de/149951106/2024_02_Response_Consultation_Art_35_Guidelines.pdf> accessed June 1, 2025.

¹⁴ Buri and van Hoboken (n 8) 33.

¹⁵ Anu Bradford, “Europe’s Digital Constitution” (Verfassungsblog, September 6, 2023) <<https://verfassungsblog.de/europes-digital-constitution/>> accessed June 1, 2025.

¹⁶ Buri and van Hoboken (n 8) 33; Florence G’sell, “The Digital Services Act: A General Assessment,” *Content Regulation in the European Union: The Digital Services Act*, vol 1 (Verein für Recht und Digitalisierung eV Institute for Digital Law Trier (IRDIT) 2023) 114.

¹⁷ Buri and van Hoboken (n 8) 33; Global Network Initiative and Digital Trust and Safety Partnership, “Implementing Risk Assessments under the Digital Services Act” (2023) 5; G’sell (n 16) 114.

¹⁸ see Buri and van Hoboken (n 8); G’sell (n 16); Leerssen (n 9); Alessandro Mantelero, “Fundamental Rights Impact Assessments in the DSA” (Verfassungsblog, November 1, 2022) <<https://verfassungsblog.de/dsa-impact-assessment/>> accessed June 1, 2025.

¹⁹ see AlgorithmWatch, “New Project: Auditing Algorithms for Systemic Risks” (AlgorithmWatch, January 19, 2023) <<https://algorithmwatch.org/en/new-project-auditing-algorithms-for-systemic-risks/>> accessed June 1, 2025.

²⁰ Sally Broughton Micova and Andrea Calef, “Elements for Effective Systemic Risk Assessment Under the DSA” (Centre on Regulation in Europe 2023) 9; e.g., in the financial sector: Regulation (EU) No 1092/2010 of the European Parliament and of the Council of 24 November 2010 on European Union macro-prudential oversight of the financial system and establishing a European Systemic Risk Board [2010] OJ L 331/1; e.g., in the healthcare sector: Jan Krzysztof Solarz and Krzysztof Waliszewski, “Holistic Framework for COVID-19 Pandemic as Systemic Risk” (2020) Volume XXIII European Research Studies Journal 340; Desheng Dash Wu, Jade Mitchell and James H Lambert, “Global Systemic Risk and Resilience for Novel Coronavirus and COVID-19” (2021) 41 Risk Analysis 701; OECD, *Emerging Risks in the 21st Century: An Agenda for Action* (Organisation for Economic Co-operation and Development 2003); e.g., in the environmental sector Klaus Lucas, Ortwin Renn and Carlo Jaeger, “Systemic Risks: Theory and Mathematical Modeling” (2018) 1 Advanced Theory and Simulations 1800051.

²¹ Efroni (n 11).

sectors, namely financial markets, to contextualise the discussions on the definition of systemic risk within the DSA.

Conventional risks are generally defined as having temporal and spatial boundaries and in terms of the correlation that exists between the likelihood of an event occurring and the size of the predicted losses.²² Rather, studies of systemic risks emerging from finance seek to capture the risks of highly interconnected systems,²³ which go beyond the typical “cause-effect” relation.²⁴ These risks often materialise following trigger events, or shocks, that initiate chain reactions across institutions or markets, and reflect the probability that the instability of a given institution will spread to other parts of the system.²⁵

In the financial sector, systemic risks can manifest in two primary ways.²⁶ First, they can arise through the amplification of failures across large, interconnected networks: networks can increase the likelihood that a failure in one of its sections will lead to the breakdown of significant portions of the remainder of the network.²⁷ This is in line with the definition that is provided in Regulation 1092/2010 of systemic risks, namely “risk of disruption in the financial system with the potential to have serious negative consequences for the internal market and the real economy”.²⁸ Second, due to the societal reliance on these networks, any failure could potentially disrupt critical social functions.²⁹ For instance, if a critical system like healthcare or energy supply were to fail, it could have significant repercussions for other aspects of society, causing further harm.³⁰ Hence, a crucial factor lies in delineating the system’s boundaries and identifying the point at which the risk of harm transitions from companies to the public.³¹

Within the DSA, the characteristics of financial markets, interconnected systems and especially societal reliance on these networks, are not necessarily mentioned but implied within Recitals 80 to 83 of the DSA. Within those Recitals, mirroring Article 34(1) DSA and as introduced above, four categories of systemic risks are mentioned. First, the societal reliance on VLOPs and VLOSEs amplifies the possibility for spreading of *illegal content* (e.g., child sexual abuse material, illegal speech, services, and products). Second, the design of the systems impact how *fundamental rights*, such as freedom of expression, information, media pluralism, can be exercised; a risk lies with design

practices that either chill the ability to exercise those rights or through manipulative design lead to addictive behaviours – and especially those of younger people.³² Third, VLOPs and VLOSEs have an effect – due to their central role in how we communicate and engage – on *democratic processes and civic discourse*. Last, the DSA addresses the risk of negative effects on the protection of *public health*, minors, and minorities in particular due to the spread of coordinated disinformation campaigns.

Article 34 DSA describes how systemic risks stem in practice from the design, functioning and use of VLOPs’ and VLOSEs’ services and related systems. It has been argued that this formulation and the focus on the “systems” may encourage overlooking external changes that may have an impact on already implemented functionalities.³³ Platforms should also assess their impact beyond those systems listed in Article 34(2), and evaluate their impact on fundamental rights as key actors in the “society”.³⁴ Uncertainty has also been expressed as to the specific methods that will be employed to identify, assess, and mitigate systemic risks, and especially as to how fundamental rights assessments will be integrated in the process.³⁵

These aspects are fundamental: in the DSA, the risk governance model adopted is based on self-assessment, and this is usually characterised by an internal perspective on potential side-effects.³⁶ These concerns remain valid after the first round of risk assessment and audit reports, released at the end of November 2024. Questions remain regarding the effectiveness of systemic risks assessments in practice; although these could represent a starting point in improving platforms’ transparency and accountability, scholars fear systemic risk management might end up becoming a performative “public relations exercise”.³⁷

The definitions and discussions within the DSA resemble the ones we see within the financial sector regulation.³⁸ User numbers, which define whether an intermediary can be qualified as “very large,” serve as representative for both the number of individuals that might be affected by the harms, and for the extent to which a service functions as a public space impacting the real world.³⁹ In this sense, Broughton Micova and

²² Lucas, Renn and Jaeger (n 20) 1; Erik Hollnagel, “A Tale of Two Safeties” (2013) 4 Nuclear Safety and Simulation 1, 2; OECD (n 20) 32.

²³ Broughton Micova (n 6) 11; Dean Curran, “Surveillance Capitalism and Systemic Digital Risk: The Imperative to Collect and Connect and the Risks of Interconnectedness” (2023) 10 Big Data & Society 6; George G Kaufman and Kenneth E Scott, “What Is Systemic Risk, and Do Bank Regulators Retard or Contribute to It?” (2003) 7 The Independent Review 371, 372.

²⁴ Terje Aven and Ortwin Renn, “Some Foundational Issues Related to Risk Governance and Different Types of Risks” (2020) 23 Journal of Risk Research 1121, 2; Hollnagel (n 22) 2.

²⁵ Broughton Micova and Calef (n 20) 26–27; Broughton Micova (n 6) 11; Curran (n 23) 6; S Poledna and others, “Systemic Risk Emerging from Interconnections: The Case of Financial Systems” (New Approaches to Economic Challenges, OECD Publishing 2020) 124.

²⁶ Curran (n 23) 6.

²⁷ *ibid.*

²⁸ Article 2(5) Regulation (EU) No 1092/2010 of the European Parliament and of the Council of 24 November 2010 on European Union macro-prudential oversight of the financial system and establishing a European Systemic Risk Board [2010] OJ L 331/1.

²⁹ Curran (n 23) 6; Lucas, Renn and Jaeger (n 20) 2.

³⁰ Research Institute for Sustainability Helmholtz Centre Potsdam, “Systemic Risks” (May 18, 2021) <<https://www.rifs-potsdam.de/en/output/dossiers/systemic-risks>> accessed June 1, 2025.

³¹ Broughton Micova (n 6) 11.

³² Yao Qin, Bahiyah Omar and Alessandro Musetti, “The Addiction Behavior of Short-Form Video App TikTok: The Information Quality and System Quality Perspective” (2022) 13 Frontiers in Psychology 12–13; Xiongfei Cao and others, “Exploring the Mechanism of Social Media Addiction: An Empirical Study from WeChat Users” (2020) 30 Internet Research 1305, 1318.

³³ Mantelero (n 18).

³⁴ *ibid.* See also José van Dijk, Thomas Poell and Martijn de Waal (eds), *The Platform Society* (1st edn, OUP 2018).

³⁵ Jordi Calvet-Bademunt and Joan Barata, “The European Commission’s Approach to DSA Systemic Risk Is Concerning for Freedom of Expression” (*Tech Policy Press*, October 30, 2023) <<https://techpolicy.press/the-european-commissions-approach-to-dsa-systemic-risk-is-concerning-for-freedom-of-expression>> accessed June 1, 2025.

³⁶ Mantelero (n 18).

³⁷ Mark Scott, “5 Things to Know about the Digital Services Act’s First Risk Assessments and Audits” (*Tech Policy Press*, December 11, 2024) <<https://techpolicy.press/5-things-to-know-about-the-digital-services-acts-first-risk-assessments-and-audits>> accessed June 1, 2025; AlgorithmWatch, “DSA: Erste Risikobewertungsberichte über systemische Risiken von großen Online-Plattformen lassen viele Fragen offen” *AlgorithmWatch* (November 29, 2024) <<https://algorithmwatch.org/de/dsa-risikobewertungsberichte/>> accessed June 1, 2025; Peter Chapman, “Advancing Platform Accountability: The Promise and Perils of DSA Risk Assessments” (*Tech Policy Press*, January 9, 2025) <<https://techpolicy.press/advancing-platform-accountability-the-promise-and-perils-of-dsa-risk-assessments>> accessed June 1, 2025; Tim Bernard, “Reading the Systemic Risk Assessments for Major Speech Platforms: Notes and Observations” (*Tech Policy Press*, December 20, 2024) <<https://techpolicy.press/reading-the-systemic-risk-assessments-for-major-speech-platforms-notes-and-observations>> accessed June 1, 2025.

³⁸ Stephen Schwarcz, “Systematic Regulation of Systemic Risk” (2019) 1 Wisconsin Law Review 1, 2–3.

³⁹ Broughton Micova and Calef (n 20) 15.

Calef advance that risks qualify as systemic if they hold the potential to inflict widespread harm on individuals or societal systems that are indispensable to the effective operation of governance and society.⁴⁰ For example, a temporary outage of Amazon Web Services in the United States affected various industries, from airlines to fast food, and also disrupted access to popular platforms like Tinder, and Netflix.⁴¹ Moreover, there is emerging evidence of digital platforms' severe implications for personal and public health,⁴² or privacy,⁴³ as well as political violence and criminal activity.⁴⁴ However, while it may be relatively simple to pinpoint the actors and sources of risk in financial markets, applying this concept to societal systems raises two main obstacles,⁴⁵ primarily because the DSA implementation process must be constantly refined to ensure effective harm reduction.⁴⁶

While generally, systemic risks increase the likelihood that a failure in one section of the network will lead to the breakdown of remaining parts of the network, characterising the digital ecosystem as the "location of failure" does not fully relate to the DSA concept of systemic risk.⁴⁷ In the context of VLOPs and VLOSEs, the systemic nature of risk pertains more to the repercussions on broader society resulting from user behaviours and service characteristics,⁴⁸ rather than the survival or collapse of platform ecosystems. This is further evidenced by the fact that failures within the financial sector have triggered financial crises, whereas none of the digital platforms have experienced a complete failure so far.⁴⁹ Moreover, the financial literature focuses on "trigger events" or "shocks" as sources of systemic risks.⁵⁰ The structure of financial players and the market are rather seen as factors influencing how these shocks spread.⁵¹ This is a fundamentally different approach compared to the one undertaken in the DSA, where the design and characteristics of VLOPs and VLOSEs are recognised as sources of risk.⁵² The notion of shocks is also limiting in light of the tendency of systemic negative effects to be the result of an accumulation of several minor incidents of harm propagated across VLOP or VLOSE networks.⁵³

3. Methodology

3.1. Scoping review

Scoping reviews use a systematic and iterative methodology to characterise and synthesise an existing or emerging body of literature on a given topic.⁵⁴ They map the breadth and depth of a research field to identify the state of existing literature and any possible gaps.⁵⁵

The aim of our scoping review in this paper is to comprehensively analyse the existing literature and case law on systemic risks to fundamental rights. To understand the architectural elements intrinsic to the definition of systemic risks, we map a body of literature that spans beyond the legal-doctrinal field of research, and that is in part antecedent to the entry into force of the DSA. Even if the DSA and this scholarship are not strictly correlated, the latter can provide practical guidance of when and how risks to fundamental rights have a "systemic" dimension. Moreover, it will highlight gaps and areas where more research might be needed. As highlighted by previous research and confirmed by our results in this paper, the application of systemic risks to the digital environment is an emerging topic, and this could affect the quantity and quality of the existing studies at the time of writing.

Our review process was based on the PRISMA extension for scoping reviews and the five-step methodological framework of Arksey and O'Malley (2005): the identification of the research question, the identification of relevant studies, the selection of studies, the charting of the studies, and the reporting of the results.⁵⁶ After conducting the scoping review, the results were structured into themes in accordance with Braun and Clarke (2006)'s methodological framework: familiarising with the data, generating initial codes, searching for themes, reviewing themes, defining and naming themes, and producing the report.⁵⁷

3.2. Gathering of literature and case law

Relevant studies were scoped in the Web of Science (WoS) and Scopus databases.⁵⁸ The search for literature was carried out in November 2025 through a keyword search. We selected the categories "social sciences" in Scopus and "law" in WoS.

In line with the view of industry stakeholders, academics, and experts from civil society – who identified that a risk can be systemic when it has a broad impact on systems, or when the risk is caused or

⁴⁰ *ibid* 13.

⁴¹ Lindsay Sain Jones and Tim Samples, "On the Systemic Importance of Digital Platforms" (2023) 25 University of Pennsylvania Journal of Business Law 141, 160–61.

⁴² Paul E Weigle and Reem MA Shafi, "Social Media and Youth Mental Health" (2024) 26 Current Psychiatry Reports 1, 2; Luca Braghieri, Ro'ee Levy and Alexey Makarin, "Social Media and Mental Health" (2022) 112 American Economic Review 3660, 3689.

⁴³ Karl van der Schyff, Stephen Flowerday and Steven Furnell, "Duplicitous Social Media and Data Surveillance: An Evaluation of Privacy Risk" (2020) 94 Computers & Security 101822, 11.

⁴⁴ Dominik Bär, Nicolas Pröllochs and Stefan Feuerriegel, "New Threats to Society from Free-Speech Social Media Platforms" (2023) 66 Communications of the ACM 37, 37; Rebecca Hamilton, "Platform-Enabled Crimes: Pluralizing Accountability When Social Media Companies Enable Perpetrators to Commit Atrocities" (2022) 63 Boston College Law Review 1349, 1418.

⁴⁵ Broughton Micova and Calef (n 20) 42.

⁴⁶ Sally Broughton Micova and Daniel Schnurr, "Systemic Risk in Digital Services: Benchmarks for Evaluating the Management of Risks to Electoral Processes" (Centre on Regulation in Europe 2024) 8.

⁴⁷ Broughton Micova and Calef (n 20) 51.

⁴⁸ José van Dijk, David Nieborg and Thomas Poell, "Reframing Platform Power" (2019) 8 Internet Policy Review 2–3; Panos Constantinides, Ola Hendridsson and Geoffrey G Parker, "Introduction—Platforms and Infrastructures in the Digital Age" (2018) 29 Information Systems Research 381, 14.

⁴⁹ Sain Jones and Samples (n 41) 151; Nizan Packin, "Too-Big-To-Fail 2.0? Digital Service Providers" (2018) 93 Indiana Law Journal 1260.

⁵⁰ Broughton Micova and Calef (n 20) 51; Kaufman and Scott (n 23) 372–73.

⁵¹ Broughton Micova and Calef (n 20) 51.

⁵² Recital 79 DSA; Broughton Micova and Calef (n 20) 50–51.

⁵³ Broughton Micova and Calef (n 20) 51.

⁵⁴ Susanne Mak and Alik Thomas, "Steps for Conducting a Scoping Review" (2022) 14 Journal of Graduate Medical Education 565, 565.

⁵⁵ Hilary Arksey and Lisa O'Malley, "Scoping Studies: Towards a Methodological Framework" [2005] International Journal of Social Research Methodology 21–22; Mak and Thomas (n 54) 565; Mai T Pham and others, "A Scoping Review of Scoping Reviews: Advancing the Approach and Enhancing the Consistency" (2014) 5 Research Synthesis Methods 371, 371–72.

⁵⁶ see Andrea C Tricco and others, "PRISMA Extension for Scoping Reviews (PRISMA-ScR): Checklist and Explanation" (2018) 169 Annals of Internal Medicine 467; Arksey and O'Malley (n 55).

⁵⁷ Virginia Braun and Victoria Clarke, "Using Thematic Analysis in Psychology" (2006) 3 Qualitative Research in Psychology 77, 87; see, also: Virginia Braun and others, "Thematic Analysis," *Handbook of Research Methods in Health Social Sciences* (Springer, Singapore 2019); Virginia Braun and Victoria Clarke, *Thematic Analysis: A Practical Guide* (SAGE 2022).

⁵⁸ see Michael Gusenbauer and Neal R Haddaway, "Which Academic Search Systems Are Suitable for Systematic Reviews or Meta-Analyses? Evaluating Retrieval Qualities of Google Scholar, PubMed, and 26 Other Resources" (2020) 11 Research Synthesis Methods 181; Raminta Pranckutė, "Web of Science (WoS) and Scopus: The Titans of Bibliographic Information in Today's Academic World" (2021) 9 Publications <<https://www.mdpi.com/2304-6775/9/1/12>>.

Table 1
Search query keyword strategy in the scoping review of literature.

Search terms used		
“systemic” AND “risk” AND (“digital” OR “online” OR “internet” OR “platform” OR “intermediary service provider” OR “social media” OR “social network” OR “technolog*”)		
Database results		
Database	Date accessed	Number of results
WoS	04/11/2025	281
Scopus	04/11/2025	946
Total		1227

Table 2
Search query keyword search strategy in the scoping review of judgements.

Search terms used		
“systemic” AND “risk” AND (“digital” OR “online” OR “internet” OR “platform” OR “intermediary service provider” OR “social media” OR “social network” OR “technolog*”)		
Database results		
Database	Date accessed	Number of results
Curia	25/11/2025	148
WestLaw Classic	25/11/2025	119
Total		267

Table 3
Exclusion criteria in the screening of abstracts and titles.

Exclusion Criteria	
1	The language of the paper is not English.
2	The record is a Master Thesis or a full edited book.
3	The abstract or title is not related to the digital environment.
4	The abstract or title does not focus on systemic risks for fundamental rights.

aggravated by a system –⁵⁹ the search query was not limited to studies using “systemic risk” in combination, but included: “systemic” AND “risk” AND (“digital” OR “online” OR “internet” OR “platform” OR “intermediary service provider” OR “social media” OR “social network” OR “technolog*”). We did not restrict the language of the results until the screening stage. The search strategy and the number of results for each database is shown in [Table 1](#).

The case law search was carried out in November 2025 on Curia and WestLaw Classic. On the Curia database, the keywords were searched for “text,” and on WestLaw Classic for “EU law” and “case law”.

Similarly to the literature search, the search query included the following search terms: “systemic” AND “risk” AND (“digital” OR “online” OR “internet” OR “platform” OR “intermediary service provider” OR “social media” OR “social network” OR “technolog*”). The number of results for each database is shown in [Table 2](#).

3.3. Selection of relevant literature and case law

Upon completion of the literature search, information on the results’ author(s), year of publication, language, titles, journal, and document type were collected into an Excel Sheet. Duplicates ($n = 56$) were manually removed before the screening process. We screened a total of 1171 titles and abstracts for exclusion according to the criteria listed in [Table 3](#): All English and literature sources that address systemic risks in the digital (or synonymous) environment were included in the scoping review if systemic risks are dealt with in connection with fundamental rights or freedoms. We did not include Master Thesis or full edited

Table 4
Exclusion criteria in the screening of case facts and outcomes.

Exclusion Criteria	
1	The record is not a judgment of the Court of Justice of the European Union.
2	The case facts are not related to the digital environment
3	The case facts do not focus on systemic risks for fundamental rights.

books. The screening was independently performed by two of the authors. In case of disagreement, the third author was consulted to reach a consensus.

The full text of the remaining records ($n = 71$) was analysed for inclusion. 34 records were at this stage excluded because the text did not address systemic risks, nor risks that impact or are impacted by systems ($n = 11$), or because the text did not discuss risks in the digital or technological environment ($n = 20$). The remaining papers ($n = 40$) were included in the thematic analysis.

As regards case law, we collected the case number, the European Case Law Identifier (ECLI), the date of decision and keywords. The information was again gathered in an Excel Sheet. Before the screening process, all records that were not a judgment of the Court of Justice (ECJ) or General Court (GC) were excluded ($n = 153$), and duplicates of these judgements ($n = 47$) were manually removed. The remaining records ($n = 67$) were independently screened by two of the authors for inclusion. The case law was summarised with regards to case facts and outcome, and the exclusion criteria included ([Table 4](#)).

After screening for exclusion, 5 judgements were analysed for inclusion. The judgments were read in full and 3 consequently excluded because the subject matter did not pertain to systemic risks, nor risks that impact or are impacted by systems ($n = 3$). 2 judgements were therefore ultimately included in the thematic discussion. The PRISMA flow diagramme summarising the study selection process for both the literature and case search can be found in [Fig. 1](#) in Annex I.

3.4. Thematic analysis

The reflexive thematic analysis was conducted following the framework of Braun and Clarke (2006). The text of the included records was marked with notes and ideas for the coding process. Then, an initial code was generated from the data by identifying “interesting” “features of the data”.⁶⁰ This process combined a “data-driven” and “theory-driven” approach.⁶¹ The codes largely depend on the data, but regard was given to finding information that could be useful in answering the research question. The extracts were coded manually, first by highlighting and taking notes on the margin of the texts, and later by copying extracts dealing with similar content under the same heading in a document. Given the variety of “digital environments” that the body of literature scoped deals with, the overarching context of the codes were also taken into consideration. Later, the different codes were sorted into potential themes. This phase ended with a collection of candidate themes, sub-themes, and a collection of codes extracted. The codes were then refined to form coherent patterns. For each theme, its “essence” was identified, so the central aspect that the theme captures, as well as how the theme fits in the overall story that is told about the data and in the answer to the research question. All authors contributed to the development of codes, their organization into patterns of data, and ultimately their refinement into overarching themes. The full list of records included in the analysis ([Table 6](#)), and quotations from the studies analysed ([Table 7](#)) were collected by the authors in Annex II.

⁵⁹ David Sullivan and Jason Pielemeier, “Unpacking ‘Systemic Risk’ Under the EU’s Digital Service Act” (*Tech Policy Press*, July 19, 2023) <<https://techpolicy.press/unpacking-systemic-risk-under-the-eus-digital-service-act>> accessed June 1, 2025.

⁶⁰ Braun and Clarke, “Using Thematic Analysis in Psychology” (n 57) 88.

⁶¹ *ibid*.

Table 5

Qualitative framework to classify systemic risks; divided in classes, steps, and lead questions to guide the analysis.

Class	Steps	Lead questions			
Risk identification	Step 1: Determine the origin	Quality and availability of data <i>Has there been integration of public and private sector data without adequate safeguards? Is there a lack of transparency in how the data is collected, processed or shared?</i>	Data concentration <i>Is the data centralised and monopolised by a few digital structure intermediaries? Does the platform's role as a "modern public square" impact the failure of fundamental rights? Does the data collected enable the amplification of emotional responses, outrage, or polarisation?</i>	Design and use of (all relevant) algorithmic systems (in particular, recommender and advertising systems) <i>Does the algorithmic system control the visibility of content or influence the behaviour of users? Is the system being used for surveillance or automated decision-making (e.g., automated security checks or assessments of migration applications)?</i>	Users of technologies <i>Can the platform or service be used in ways that reinforce social or cognitive bias? Is it being used to mobilise political or social movements in harmful ways? Are users (and especially vulnerable groups like children) exposed to risks such as exploitation or abuse through the platform or service?</i>
Risk identification	Step 2: Determine rights impacted (Article 34 DSA)	Any actual or foreseeable negative effects for the exercise of fundamental rights	E.g., in line with the EU Charter of Fundamental Rights, human dignity, respect for private and family life, protection of personal data, freedom of expression and information (including the freedom and pluralism of the media), right to non-discrimination, rights of the child, right to a high-level of consumer protection		
Impact of Risk	Step 3: Determine scale of impact	Impact on society <i>Does the risk affect both individuals and society as a whole? Could the risk contribute to changes in the structure or functioning of society? Could this risk lead to the erosion of fundamental rights? Does the risk create challenges in legal frameworks (which would make it difficult to assign liability or find remedies)?</i>	Impact on individuals <i>Does the risk weaken fundamental rights of the individual (right to privacy, freedom of expression, right to non-discrimination, right to financial inclusion, human dignity, consumer protection)? Does the risk disproportionately affect vulnerable populations (children, minorities, individuals with disabilities, elderly)? Could this risk perpetuate existing biases or inequalities? Could this risk lead to discrimination or exclusion? Could individuals lose control over their personal data?</i>	Impact on economy <i>Could the risk have an impact on the economic or financial stability? Does the risk involve personal data markets as financial assets? Could the risk lead to financial discrimination or financial exclusion in the economy? Does the risk create an imbalance of power between consumers and large tech companies?</i>	
Impact of Risk	Step 4: Determine the remediability of the impact	Would mitigating this risk require system-wide interventions rather than individual solutions?	Does the risk have self-perpetuating or amplifying effects that make it difficult to mitigate?	Are there clear legal remedies available to address the consequences of the risk?	Could the risk lead to long-lasting restrictions on fundamental rights?
Probability of Risk	Step 5: Determine the probability	Is the system deeply integrated in society?	Does regulatory protection exist? Are there laws or policies that effectively mitigate the risk?	Are there known vulnerabilities? Is there a history of data misuse or surveillance? Are there patterns of discrimination? Is the risk associated with known biases?	Is this behaviour financially incentivised (for example, data monetisation)?

3.5. Limitations

While the search for relevant judgements was conducted on two specialised case law databases, we limited the scoping research for literature to Scopus and WoS to focus on articles published in renowned publication venues in law, economics and other disciplines. Although our literature search resulted in 1171 titles, the thematic analysis only included 40. The search was conducted with broad keywords in the hope of capturing as many risks to fundamental rights as possible, without being limited by the ones mentioned in Article 34 DSA. By conducting the search on more law or economics specialised databases, we could have captured a broader class of definitions of systemic risks. Moreover, as many of these titles discuss primarily "system" instead of "systemic," referring therefore to systemic risks as "risks to systems," another search strategy could have included "system" within the keywords to capture a broader class of risks. Additionally, as the screening for inclusion was primarily decided based on the abstracts of the titles, we perhaps have missed contributions because fundamental rights were not mentioned. A future search could also expand the scope to include the database of the European Court of Human Rights in addition to the Court of Justice of the European Union.

4. Results

4.1. Quantitative results

The scoping review analysed a total of 40 publications: 36 journal articles, 3 book chapters, and 1 book. The majority of these studies employed qualitative research methods ($n = 33$, 82.5 %). Other studies used mixed methods ($n = 6$, 15 %), incorporating survey data or statistics⁶² or interviews⁶³ alongside qualitative research. Only one study

⁶² see Irina Aseeva and Vladimir Budanov, "Digitalization: Potential Risks for Civil Society" (2020) 186 *Economic Annals-XXI* 36; Sasidaran Gopalan, Bhavya Gupta and Ramkishan Rajan, "Financial Globalisation in ASEAN +3: Navigating the Financial Trilemma" (2023) 14 *Global Policy* 464; FangBing Zhu and Zongyu Song, "Systematic Regulation of Personal Information Rights in the Era of Big Data" (2022) 12 *SAGE Open* 21582440211067529.

⁶³ see Sarah Hill, "Navigating Visibility and Risk: Disabled Young Women's Self-Presentation Practices on Social Media" (2023) 33 *Journal of Gender Studies*; Kevin Quigley and Jeffrey Roy, "Cyber-Security and Risk Management in an Interoperable World An Examination of Governmental Action in North America" (2012) 30 *Social Science Computer Review* 83.

(2.5 %) relied solely on quantitative research methods.⁶⁴ This indicates a preference for qualitative research in this field.

The publications in the scoping review span 2012–2025, indicating that the academic interest on systemic risks to fundamental rights in the digital environment is relatively new. Since 2020, there has been a growing interest in the topic (with 35 out of the 40 analysed publications being published between 2020 and 2025). The fact that 12 included studies were published from 2025 suggests that systemic risk to fundamental rights is emerging as a distinct field of literature.

Within the studies in review, a smaller portion originated from the legal field ($n = 17$, 42.5 %). The majority of studies originated from social sciences or present an interdisciplinary background ($n = 23$, 57.5 %). It is interesting to observe that “systemic risks” are used in combination in 18 studies (45 %), while the remaining 22 discussed “risks” as affecting or aggravating a system (55 %). 3 of the 40 studies develop the notion of systemic risks in connection to the financial market (7.5 %), while 10 studies discuss systemic risks under the Digital Services Act (25 %). Although not strictly the focus of the review, we highlight that several of the studies discuss systemic risks in connection to the obligations of the Artificial Intelligence Act, whose scope also includes risks to fundamental rights ($n = 7$, 17.5 %).

The body of literature discusses risks that impact society in relation to several fundamental rights. Most studies address multiple fundamental rights simultaneously ($n = 25$, 62.5 %), while only a minority focuses on the discussion of a single right ($n = 15$, 37.5 %). The right to privacy ($n = 18$, 45 %) and the right to non-discrimination ($n = 16$, 40 %) are the most frequently addressed within the studies, while access to finance ($n = 3$, 7.5 %), children’s rights ($n = 9$, 22.5 %) and consumer protection ($n = 9$, 22.5 %) the least. Additional rights that also emerged in multiple studies are freedom of expression ($n = 12$, 30 %), human dignity ($n = 10$, 25 %).

The scoping review also includes recent court developments, notably two 2025 General Court of the EU decisions that directly addressed systemic risks: Case T-348/23 (Zalando v Commission) and Case T-367/23 (Amazon EU v Commission).

4.2. Qualitative results: when is a risk to fundamental rights “systemic”?

From the thematic analysis, we constructed three main themes around the question of when a “risk” triggers the threshold of “systemic”: 1) structural characteristics of systemic risks; 2) systemic by techno-individual-societal sources; 3) systemic by societal consequences.

As a prerequisite, in order to fall under the scope of DSA systemic risks to fundamental rights, the literature reminds four minimum legal requirements. First, systemic risks are uniquely associated with **VLOPs and VLOSEs**: entities that meet specific legislative thresholds for average monthly active recipients.⁶⁵ Because of their scale, these providers pose societal and economic risks that differ significantly from those of smaller platforms, and they have been formally designated by the Commission.⁶⁶ Second, the risk must be intrinsically **linked to the design, functioning, and use** of the service.⁶⁷ In this context, “design” refers to architectural choices of the VLOPSE, while “functioning” encompasses the platform’s objective operational modes.⁶⁸ Third, the risk must fall within the **Article 34(1) DSA category**, specifically regarding

negative effects on fundamental rights.⁶⁹ Even if the VLOPSE operates independently without constituting part of a unified system – such as the financial system – this independence does not exempt them from the systemic risk obligations.⁷⁰ Designated individual platforms can still give rise to the systemic risks in Article 34(1) DSA regardless of their lack of connection with other entities.⁷¹ Finally, the risk must possess a **Union-wide dimension**, demonstrating a likelihood of affecting a significant portion of the EU population.⁷²

4.2.1. Theme 1: the structural characteristics of systemic risks

The literature recognises several structural characteristics of systemic risks to fundamental rights. At their core, systemic risks are best understood as complex, uncertain and ambiguous.⁷³ Because systemic risks stem from the entire lifecycle of a system (design, implementation and usage), it is difficult to attribute a specific outcome to a specific cause or source (*complexity*).⁷⁴ There is a lack of scientific consensus on how to manage these risks, and since they are often “recurring and structural,” their potential negative outcomes are hard to predict or prevent (*uncertainty*).⁷⁵ Last, there is often disagreement on how to identify, evaluate and justify these risks and their potential adverse effects (*ambiguity*).⁷⁶ There is no objective way to assess them through standard metrics or probability calculations. On one hand, identifying risks to fundamental rights goes beyond assessing if a right will be violated or not: interferences to rights exist on a spectrum of different degrees rather than as a binary calculation.⁷⁷ On the other hand, identifying systemic risks often involves balancing power dynamics and competing interests, but also institutional dynamics.⁷⁸ Kusche writes that “the introduction of systemic risks opens up a decision-making that,

⁶⁹ Marta Beltrán, “AI Algorithms under Scrutiny: GDPR, DSA, AI Act and CRA as Pillars for Algorithmic Security and Privacy in the European Union” (2025) 158 Computers & Security 104628, 4; Rita Gsenger and Marie-Therese Sekwenz, “The Digital Services Act: Online Risks, Transparency and Data Access” in Rita Gsenger and Marie-Therese Sekwenz (eds), *Digital Decade* (Nomos Verlagsgesellschaft mbH & Co KG 2025) 130–31 <<https://www.inlibra.com/de/document/view/detail/uuid/4a1e0d0b-32d6-366d-a594-f7babea0f79>> accessed June 1, 2025; Afroditi Papathanasopoulou, “From Professional Fact-Checkers to the Crowd: Can Meta’s Community Notes Survive the Digital Services Act (DSA)?” [2025] Information & Communications Technology Law 1, 4; also, *Case T-348/23 Zalando v Commission* [2025] para 132; *Case T-367/23 Amazon EU v Commission* [2025] paras 65–68.

⁷⁰ *Amazon EU v Commission* (n 69) paras 65, 70.

⁷¹ *ibid* paras 65, 70.

⁷² Kaesling and Wolf (n 65) 120–21; Gergely Ferenc Lendvai, “Taming the Titans? – Digital Constitutionalism and the Digital Services Act” (2024) 17 ESSACHESS – Journal for Communication Studies 169, 179; *Amazon EU v Commission* (n 69) para 70.

⁷³ Lucas M Haitsma, “Addressing Discrimination in Algorithmic Profiling: Examining Risk Governance in Dutch Public Social Security Agencies” (2025) 27 European Journal of Social Security 191, 192–93.

⁷⁴ *ibid* 195.

⁷⁵ *ibid*.

⁷⁶ *ibid*.

⁷⁷ Gianclaudio Malgieri and Cristiana Santos, “Assessing the (Severity of) Impacts on Fundamental Rights” (2025) 56 Computer Law & Security Review 106113, 5.

⁷⁸ Wendy L Currie and others, “Rethinking Technology Regulation in the Age of AI Risks” (2025) 40 Journal of Information Technology 236, 238; Ronan Ó Fathaigh, Doris Buijs and Joris Van Hoboken, “The Regulation of Disinformation Under the Digital Services Act” (2025) 13 Media and Communication 9615, 9; Andrea Palumbo, “A Medley of Public and Private Power in DSA Content Moderation for Harmful but Legal Content: An Account of Transparency, Accountability and Redress Challenges” (2024) 15 JIPITEC – Journal of Intellectual Property, Information Technology and E-Commerce Law 256.

⁶⁴ see Eliane Rösli, Selen Bozkurt and Tina Hernandez-Boussard, “Peeking into a Black Box, the Fairness and Generalizability of a MIMIC-III Benchmarking Model” (2022) 9 Scientific Data 24.

⁶⁵ Katharina Kaesling and Annegret Wolf, “Sustainability and Risk Management under the Digital Services Act: A Touchstone for the Interpretation of ‘Systemic Risks’” (2025) 74 GRUR International 119, 119.

⁶⁶ *ibid*.

⁶⁷ *ibid* 121.

⁶⁸ *ibid* 128.

although not necessarily arbitrary, has nothing to do with the estimation of risk and everything to do with negotiations between interests that hugely differ in terms of power.⁷⁹ Risks to fundamental rights are often less tangible, and as a consequence more contestable.⁸⁰

4.2.2. Theme 2: the sources of systemic risks to fundamental rights

Systemic risks to fundamental rights can be identified by looking at the sources of risk. These sources can be grouped under a) quality and availability of data, b) data concentration, c) design and deployment of algorithms, and d) users of the technologies. These themes are holistic and include both internal and external sources of systemic risks.⁸¹

Quality and availability of data. The proliferation of the internet has resulted in substantial modifications to the methods and scope of collection of personal data.⁸² Information in the data market is of increasing quality and interconnectedness.⁸³ This is leading to society experiencing an integration of data across and between the public and private sector. For instance, the data collected by private enterprises for commercial purposes and by law enforcement for investigative purposes are often combined,⁸⁴ and in numerous domains, digital or electronic identification has become a standard practice.⁸⁵ Data on reading interests, Internet collections, social media links, or travel patterns, could therefore potentially be used to reveal those who are more likely to protest against the government.⁸⁶ This is particularly problematic if, for instance, these large amounts of behavioural, demographic and socioeconomic data collected are relied on in AI-driven legislative systems.⁸⁷

Data concentration. While data is obtained, processed, and exchanged in innovative ways, at the same time it is also increasingly concentrated in the hands of few technological enterprises.⁸⁸ These platforms function as “digital infrastructures that enable two or more groups to interact,” in their capacity of intermediaries, gather, analyse, and monopolise data.⁸⁹ As platforms become the primary venue for public life and politics, these have transformed into a modern “public

square” for social, commercial and political engagement.⁹⁰ Platforms have proven crucial in mobilising social and political movements.⁹¹ For instance, in the Arab Spring and the protests following George Floyd’s killing, social media was able to amplify group-based emotions and intensified a collective response to a perceived injustice.⁹² This is how, during the pandemic, activists using social media effectively were able to leverage the intense shared emotions to generate broader support of protest actions.⁹³ Also due to the growth in storage and analytical power of platforms, future applications of the data collected become inherently hard to predict.⁹⁴ Moreover, because algorithms are trained on historical data, these might encore and amplify existing biases related to, for instance, gender or class – replicating and entrenching systemic inequalities.⁹⁵

Design and deployment of algorithms. Algorithms are central. Actions are increasingly converted into computer-programmed instructions for stronger computational power.⁹⁶ For platforms such as social media, algorithmic changes deeply affect how and what content that can be shared.⁹⁷ Through algorithms, platform can easily quantify users’ social behaviour, and in this way predict and influence their attitudes.⁹⁸ And options to identify correlations and construct new profiles appear endless.⁹⁹

Users of technologies. The context of use of the different technologies has an impact on their risk profile.¹⁰⁰ For instance, this has been observed in the use of platforms for video-conferencing interviews. “While video conference interviews can be both convenient and cost effective, they may also exacerbate existing biases and introduce new sources of bias into the recruitment process”.¹⁰¹ Given that “communication delays of just over 1 s can cause conversation partners to be perceived as less extroverted, conscientious, and attentive,” applicants with a slower internet connection or a lower-quality camera may be penalised.¹⁰² On social media platforms, peer-to-peer harms are increasing. For instance, children engaging in cyberbullying, or sharing or (AI-)generating online child sexual abuse material.¹⁰³

4.2.3. Theme 3: the societal impact of systemic risks to fundamental rights

Systemic risks to fundamental rights impact individuals and society in different ways. Literature recognises challenges to a) individuals, b) society and c) the economy.

Theme 3a: Individuals

The effects of technological advances are not uniform: factors such as demographics, socioeconomic status and mental health conditions have an effect in understanding the societal impacts of risks.¹⁰⁴

⁷⁹ Isabel Kusche, “Possible Harms of Artificial Intelligence and the EU AI Act: Fundamental Rights and Risk” [2024] *Journal of Risk Research* 1, 11.

⁸⁰ Mateus Correia de Carvalho, “It Will Be What We Want It to Be: Socio-technical and Contested Systemic Risk at the Core of the EU’s Regulation of Platforms’ AI Systems” (2025) 16 *Journal of Intellectual Property, Information Technology and E-Commerce Law* 37.

⁸¹ see, Kyooeun Jang, “Peeling Back the Layers of ‘Paint on Rotten Wood’: Unraveling the Senate’s ‘Big Tech and Child Sexual Exploitation Crisis’ Hearing” (2025) 19 *International Journal of Communication* 1828, 1829.

⁸² Stefan Strauß, *Privacy and Identity in a Networked Society: Refining Privacy Impact Assessment* (1st edn, Routledge 2019) 87; Zhu and Song (n 62) 1.

⁸³ Aisling McMahon, Alena Buyx and Barbara Prainsack, “Big Data Governance Needs More Collective Responsibility: The Role of Harm Mitigation in the Governance of Data Use in Medicine and Beyond” (2020) 28 *Medical Law Review* 155, 156.

⁸⁴ Andrew Guthrie Ferguson, “Big Data Surveillance: The Convergence of Big Data and Law Enforcement” in David Gray and Stephen Henderson (eds), *The Cambridge Handbook of Surveillance Law* (Cambridge University Press 2017) 171; Gopalan, Gupta and Rajan (n 62) 473; Joseph Lee, “Access to Finance for Artificial Intelligence Regulation in the Financial Services Industry” (2020) 21 *European Business Organization Law Review* 731, 748–49.

⁸⁵ Strauß (n 82) 93.

⁸⁶ Ferguson (n 84) 178.

⁸⁷ Firdaus Arifin and others, “Designing an AI-Driven Legislation Framework to Improve Indonesia’s Law-Making Transparency and Public Participation” (2025) 14 *Yustisia Jurnal Hukum* 152, 164.

⁸⁸ Dirk Zetsche and others, “The Evolution and Future of Data-Driven Finance in the EU” (2020) 57 *Common Market Law Review* 355.

⁸⁹ Simon Elias Bibri, “The Metaverse as a Virtual Model of Platform Urbanism: Its Converging AIoT, XReality, Neurotech, and Nanobiotech and Their Applications, Challenges, and Risks” (2023) 6 *Smart Cities* 1345, 1350; Paula Helm and Sandra Seubert, “Normative Paradoxes of Privacy: Literacy and Choice in Platform Societies” (2020) 18 *Surveillance & Society* 185, 186–87.

⁹⁰ Papathanasopoulou (n 69) 1; Lendvai (n 72) 170.

⁹¹ Peter Grant and Heather Smith, “Activism in the Time of COVID-19” (2021) 24 *Group Processes & Intergroup Relations* 297, 301; Andrea Lucchi, “Freedom of Expression and the Right of Access to the Internet: A New Fundamental Right?” in Monroe Price, Stefaan Verhulst and Libby Morgan (eds), *Routledge Handbook of Media Law* (1st edn, Routledge 2013) 158.

⁹² Grant and Smith (n 91) 301; also, on amplification: Palumbo (n 78) 250.

⁹³ Grant and Smith (n 91) 301.

⁹⁴ Helm and Seubert (n 89) 188.

⁹⁵ Arifin and others (n 87) 164.

⁹⁶ Bibri (n 89) 1351.

⁹⁷ Sophie Bishop, “Influencer Management Tools: Algorithmic Cultures, Brand Safety, and Bias” (2021) 7 *Social Media + Society* 205630512110030, 1.

⁹⁸ Bibri (n 89) 1351.

⁹⁹ Helm and Seubert (n 89) 188.

¹⁰⁰ Currie and others (n 78) 238.

¹⁰¹ Jyothi Marbin, Y-Vonne Hutchinson and Sarah Schaeffer, “Avoiding the Virtual Pitfall: Identifying and Mitigating Biases in Graduate Medical Education Videoconference Interviews” (2021) 96 *Academic Medicine: Journal of the Association of American Medical Colleges* 1120, 1120.

¹⁰² *ibid* 1121.

¹⁰³ Jang (n 81) 1832.

¹⁰⁴ *ibid* 1831.

The use of algorithms in the provision of services raises perpetuating inequalities and societal bias towards traditionally marginalised groups, thereby challenging the framework of non-discrimination laws.¹⁰⁵ Bishop notes that engineering practices and training datasets exhibit a long-standing history of discriminatory decision-making, and this is perpetuating patterns of bias in predictive modelling.¹⁰⁶ Predictive analytics can lead to discrimination by excluding individuals deemed risky from services, fostering stigmas, or economic disadvantages without legal recourse due to the opacity of data use and analysis.¹⁰⁷ Algorithmic biases encoded in platform design have far-reaching consequences, leading to poor consumer ratings, denial of mortgages, provision of welfare subsidies, difficulties in employment, and increased insurance premiums.¹⁰⁸ In evaluating the brand safety of influencers, biases penalise the work of influencers belonging to minority communities; for instance, classifying the use of words such as “queer” as brand unsafe in a decontextualised way results in lower scores for creators who use this word simply because they belong to the LGBTQ+ community.¹⁰⁹ Similar obstacles arise from the deployment of AI-driven technologies in the healthcare sector, where “models should also systematically undergo fairness assessments to break the vicious cycle of inadvertently perpetuating the systemic biases found in society and healthcare under the mantle of AI.”¹¹⁰

It is therefore clear that algorithmic discrimination often targets in particular groups who are already intersectionally marginalised in society.¹¹¹ For example, ageist, ableist and racist logics were all deeply ingrained in nursing homes prior to the introduction of algorithmic surveillance technologies, shaping the context in which these technologies now operate.¹¹² Bias is also identified in recommendation systems and engagement strategies, and it extends to algorithmic decision-making in education, employment, and insurance, where discrimination against individuals with disabilities is promoted due to differences in communication, interaction, and being.¹¹³

Efforts to create race-neutral technologies frequently fail because they overlook the broader socio-economic structures that perpetuate white supremacy rhetoric and inequality, reducing fairness to a technical problem solvable by improved data and design alone.¹¹⁴ In this regard, migrants are particularly at risk in the use of facial recognition technologies, which are prone to inaccuracies for individuals with darker skin tones, leading to exclusion from essential services.¹¹⁵

Ferguson writes that personal data on the internet is “continuously bought, sought, and repurposed by a whole network of advertisers, ad exchanges, and market researchers”.¹¹⁶ To describe how digital individual identities are construed he reports this quote by Thoman Hemnes:

“One might think of a person’s digital identity by analogy to a pointillist painting. Thousands upon thousands of tiny bits of digital information about an individual, including what we have called basic facts, sensitive facts, and transactional facts, can be assembled to form a picture of the individual: his likes, dislikes, predispositions, resources; and in fact, any facet of his personality that has had contact with the Internet.”¹¹⁷

The interconnectedness and availability of data has blurred the lines between private and public domains and personal and non-personal information: as digital information moves across socio-technical contexts, a permanent connection to the identity of an individual person can now be established.¹¹⁸ Customers, while benefiting from user-customised services and targeted marketing, have experienced leakages of personal information and a “seemingly unavoidable” invasion of their privacy.¹¹⁹ Conventional privacy frameworks, which rely heavily on individual autonomy and consent, struggle to address the systemic power of dominant digital platforms.¹²⁰ For children, this persistent tracking, or “dataveillance”, begins very early and often continues through childhood through “sharenting” and educational monitoring.¹²¹ This digital footprint can ultimately restrict their future employment and economic opportunities.¹²²

Advancements in technology also play a role in worsening the respect for human dignity. Bibri exemplifies this through the discussion of biometric technologies, which were instrumental in the repurposing of Meta, Google and Apple platforms during the Covid-19 pandemic to facilitate contact tracing. By reducing individuals to a set of “biometric parameters,” biometric technologies dehumanise individuals and undermine their dignity.¹²³

Theme 3b: Society

Across the body of literature, it is recognised that risks created by technology affect individuals but have also strong implications for society at large. Risks to fundamental rights go beyond users of the technology, and protection must encompass society at large: “Systemic risks affect everyone, not just the users, including all people who never agreed to the terms and conditions.”¹²⁴ Technologies have, after all, become intrinsic elements of society and were integrated in a wide variety of everyday life contexts.¹²⁵ While technologies have historically been systems, with such a system being defined as a “specific organisational entity constituted by a set of related elements with particular characteristics”, these advancements have transformed them into socio-technical ones.¹²⁶ The interaction of the technical system’s components, occurring both within and outside the system’s environment, ties technology and society together and determines a mutual co-shaping relationship between the two.¹²⁷ Information technology is driving society into a “stage of convergence” between physical and digital environments,¹²⁸ thereby significantly facilitating interoperability among both the private and public sectors, with for instance e-government

¹⁰⁵ Clara Berridge and Alisa Grigorovich, “Algorithmic Harms and Digital Ageism in the Use of Surveillance Technologies in Nursing Homes” (2022) 7 *Frontiers in Sociology* 4; Bishop (n 97) 5–6; Marie McAuliffe, Jenna Blower and Ana Beduschi, “Digitalization and Artificial Intelligence in Migration and Mobility: Transnational Implications of the COVID-19 Pandemic” (2021) 11 *Societies* 135, 8; Rösli, Bozkurt and Hernandez-Boussard (n 64) 9; Raphaële Xenidis, “Tuning EU Equality Law to Algorithmic Discrimination: Three Pathways to Resilience” (2020) 27 *Maastricht Journal of European and Comparative Law* 736, 2; Arifin and others (n 87) 164.

¹⁰⁶ Bishop (n 97) 5.

¹⁰⁷ McMahon, Buyx and Prainsack (n 83) 161.

¹⁰⁸ *ibid*; Saabiq Chowdhury, “Technology Is Never Neutral: Robodebt and a Human Rights Analysis of Automated Decision-Making on Welfare Recipients” (2024) 30 *Australian Journal of Human Rights* 20, 22–23.

¹⁰⁹ Bishop (n 97) 8–9.

¹¹⁰ Rösli, Bozkurt and Hernandez-Boussard (n 64) 9.

¹¹¹ Bishop (n 97) 6.

¹¹² Berridge and Grigorovich (n 105) 13.

¹¹³ *ibid* 5.

¹¹⁴ *ibid* 6; Arifin and others (n 87) 164.

¹¹⁵ McAuliffe, Blower and Beduschi (n 105) 8.

¹¹⁶ Ferguson (n 84) 173–74.

¹¹⁷ Thomas Hemnes, “The Ownership and Exploitation of Personal Identity in the New Media Age” (2012) 12 *The John Marshall Review of Intellectual Property Law* 17; Ferguson (n 84) 174.

¹¹⁸ Strauß (n 82) 6.

¹¹⁹ Zhu and Song (n 62) 1.

¹²⁰ Helm and Seubert (n 89) 187.

¹²¹ Jang (n 81) 1831–32.

¹²² *ibid* 1832.

¹²³ Bibri (n 89) 1369.

¹²⁴ Niklas Eder, “Making Systemic Risk Assessments Work: How the DSA Creates a Virtuous Loop to Address the Societal Harms of Content Moderation” (2024) 25 *German Law Journal* 1197, 1205.

¹²⁵ Aseeva and Budanov (n 62) 38; McAuliffe, Blower and Beduschi (n 105) 1; Strauß (n 82) 4; Beltrán (n 69) 1.

¹²⁶ Strauß (n 82) 15.

¹²⁷ *ibid* 16–17.

¹²⁸ *ibid* 4.

initiatives that deliver online services and enhance information sharing or public engagement.¹²⁹ Transformations of the organisation and functioning of society have enabled new forms of risks.¹³⁰ These new risks, affecting both individual rights and society as a whole, have been defined as “systemic”.¹³¹ These risks transcend binary framings of offline versus cyber harms.¹³²

The development of technology requires that legal theory and processes be continuously revised in order to effectively handle emerging challenges within broader societal and organisational contexts.¹³³ This is the only means by which protection of digital rights and digital security can be guaranteed.¹³⁴ For instance, McMahon et al. recognised the challenges of identifying appropriate legal remedies in an interconnected data market. As data is sourced from different origins, tracing this source and assigning responsibility might constitute obstacles.¹³⁵

The structure and foundations of society could be affected by employing technologies, particularly exposing children or vulnerable communities.¹³⁶ Algorithms have been used to enhance surveillance capabilities in nursing homes, in migration and mobility, but also in influencer marketing tools.¹³⁷ These systems use algorithms to automate pattern recognition, whether it be to monitor the activities of older adults to guarantee their safety, or to select influencers for brand collaborations, and machine learning techniques to process the data that is continuously collected.¹³⁸ In migration, visa and asylum applications are increasingly processed through algorithms, and security checks at the borders are often automated.¹³⁹ Nevertheless, the convergence of new technologies and evolving social media habits have made everyone vulnerable to both government and corporate monitoring.¹⁴⁰ This vulnerability stems from the massive digital footprints we generate through daily financial transactions, messages, and social networking activities.¹⁴¹ One of the possible outcomes in combining algorithmisation with the network character of a society deeply intertwined with technology is the development of a “society of total control”.¹⁴² To achieve total control, surveillance and interconnectedness of data could be leveraged in the form of social ranking, and civil society will be entrusted with the task of defending human rights such as the rights to privacy and data protection.¹⁴³

Curtailling the right to privacy has a direct link to the impairment of freedom of expression. For instance, Pavone explains that the “unrestrained and unaccountable” use of large-online surveillance has resulted in a loss of control over an individual’s data.¹⁴⁴ Since electronic channels are seen as essential to democratic systems for the exchange of

information and the advancement of human freedom, all civil and democratic rights that rely on the maintenance of privacy, and the intimacy of human behaviour are also lost without privacy.¹⁴⁵ Therefore, loss of privacy is a threat for freedom of expression, for a free and political debate, for democracy, and ultimately society.¹⁴⁶

Theme 3c: Economy

Personal information has evolved into a critical asset with substantial market value.¹⁴⁷ Market integrity benefits from greater data sharing because of an improved transparency; transparency allows for better financial stability as policymakers can make informed decisions when establishing financial policies.¹⁴⁸

Society benefits from personal data markets and the use of AI in finance because it allows for new possibilities of financial inclusion.¹⁴⁹ Lee defines financial inclusion as meaning that “individuals, regardless of their background or income, have access to useful and affordable financial products and services,” including “products and services, as well as transactions and payment systems, and the use of financial technology”.¹⁵⁰ Financial inclusion should be one of the top concerns in controlling the design, development, and implementation of AI since, if AI or machine learning techniques are used in the decision of providing financial services, this could have consequences for financial inclusion other than discrimination.¹⁵¹

Nevertheless, while AI in finance can make personal recommendations on the basis of specific factors relating to the consumer’s life or situation, in this relationship, customers are vulnerable because they cannot be deemed to fully understand the advice provided.¹⁵² In the processing and collecting of information, the rights of consumers could be further impacted when the provider of a service, for example, decides to use lawfully processed information for further purposes like targeted advertising, leveraging the information asymmetry with consumers.¹⁵³

To achieve financial interests while respecting fundamental rights, infrastructure providers in the personal data market should continue to be independent, and data should be portable in the interest of financial consumers.¹⁵⁴ Moreover, as large tech companies hold vast amounts of consumer data in their services, data misuse could especially be a problem for those countries without a strong data protection regime in place.¹⁵⁵

5. Analysis

The findings of our scoping review have directly informed the development of a framework that supports determining whether and how a risk has become *systemic* under the meaning of the DSA. It operationalises the main observations drawn from the thematic analysis and aligns these with the text of the DSA.

5.1. General observations

This scoping review highlights an emerging area of research, characterised by a greater prevalence of qualitative studies over empirical ones. Moreover, the fact that studies predominantly originate from the social sciences or are interdisciplinary aligns to the general understanding of “systemic,” which before the DSA had not traditionally been

¹²⁹ Quigley and Roy (n 63) 84.

¹³⁰ Lucchi (n 91) 157; Strauß (n 82) 6; Zetzsche and others (n 88) 335.

¹³¹ Lucchi (n 91) 157.

¹³² Johnny Chan, “Reviewing and Reconceptualising Cyberbullying through a Social and Sustainable Lens” (2025) 62 *The Social Science Journal* 757, 762.

¹³³ Zhu and Song (n 62) 3; Beltrán (n 69) 1.

¹³⁴ Zhu and Song (n 62) 3.

¹³⁵ McMahon, Buyx and Prainsack (n 83) 156–57.

¹³⁶ Afroz Kaviani Johnson and Julia Sloth-Nielsen, “Child Protection, Safeguarding and the Role of the African Charter on the Rights and Welfare of the Child: Looking Back and Looking Ahead” (2020) 20 *African Human Rights Law Journal* 643, 663–64; Lucchi (n 91) 157.

¹³⁷ McAuliffe, Blower and Beduschi (n 105) 8; Bishop (n 97) 1–2; Berridge and Grigorovich (n 105) 7.

¹³⁸ McAuliffe, Blower and Beduschi (n 105) 8; Bishop (n 97) 1–2; Berridge and Grigorovich (n 105) 7.

¹³⁹ McAuliffe, Blower and Beduschi (n 105) 10.

¹⁴⁰ Thomas P Crocker, “The Fourth Amendment at Home” (2020) 96 *Indiana Law Journal* 167, 174.

¹⁴¹ *ibid.*

¹⁴² Aseeva and Budanov (n 62) 45.

¹⁴³ *ibid.* 45–46.

¹⁴⁴ Vincenzo Pavone, Elvira Santiago Gómez and David-Olivier Jaquet-Chiffelle, “A Systemic Approach to Security: Beyond the Tradeoff between Security and Liberty” (2016) 12 *Democracy and Security* 1, 15.

¹⁴⁵ *ibid.*; Lucchi (n 91) 159.

¹⁴⁶ Pavone, Santiago Gómez and Jaquet-Chiffelle (n 144) 15; Strauß (n 82) 6.

¹⁴⁷ Strauß (n 82) 137.

¹⁴⁸ Gopalan, Gupta and Rajan (n 62) 473.

¹⁴⁹ *ibid.*; Lee (n 84) 732; Zetzsche and others (n 88) 355–56.

¹⁵⁰ Lee (n 84) 732.

¹⁵¹ Gopalan, Gupta and Rajan (n 62) 473.

¹⁵² Lee (n 84) 742.

¹⁵³ Gopalan, Gupta and Rajan (n 62) 473; Strauß (n 82) 269.

¹⁵⁴ Zetzsche and others (n 88) 355–56.

¹⁵⁵ Gopalan, Gupta and Rajan (n 62) 473.

applied to online platforms or search engines. Outside of the financial sector and the specific context of the DSA, most studies do not address systemic risks as such but rather systemic risks as risks that can aggravate or impact a system. This is in line with the definition emerged from the workshop conducted in 2023 conducted by the Global Network Initiative in combination with the Digital Trust & Safety Partnership.¹⁵⁶ Nevertheless, given the prominence of this notion for the risk assessment framework of the DSA,¹⁵⁷ we see that the number of studies addressing systemic risks in the digital environment has increased, with 10 records discussing systemic risks under the DSA published between 2023 and 2025, and will foreseeably continue to do so in the future.

We see an alignment between the recitals, articles and case law of the DSA and the literature's main themes. For instance, the literature highlights how platforms are critically interconnected and serve as infrastructures that significantly influence various aspects of society. While the General Court emphasises that "interconnectedness" as understood in the financial sector is not a prerequisite to give rise to systemic risks, because the DSA "does not seek to prevent the systemic risks that [platforms] could pose in so far as they form part of a system,"¹⁵⁸ this aspect as applied to the importance and societal reliance on VLOPs and VLOSEs also emerges in Recital 75 DSA. The DSA Preamble emphasises that VLOPs and VLOSEs have become fundamental to the functioning of society "in facilitating public debate, economic transactions and the dissemination to the public of information, opinions and ideas and in influencing how recipients obtain and communicate information online."

The literature search brings in examples of how systemic risks to fundamental rights could arise from the "design, functioning, and the use of [platforms'] services" and from "potential misuses by the recipients of the service" (Recital 79 DSA), which we categorise into four primary subthemes: a) quality and availability of data, b) data concentration, c) design and deployment of algorithms, and d) users of technologies. Moreover, the literature also emphasises that VLOPs and VLOSEs "may cause societal risks, different in scope and impact from those caused by smaller platforms" (Recital 75 DSA) and that "the way they design their services [...] can cause societal concerns" (Recital 79 DSA). This leads, for instance, to the perpetuation of systemic biases and discrimination against marginalised groups, the creation of "data-veillance" especially for children, and the potential emergence of a "society of total control characterised by mass surveillance and the erosion of democratic foundations such as privacy and freedom of expression. Last, both the DSA and the literature search reference similar rights as possibly impacted by systemic risks: human dignity, freedom of expression and information, privacy and private life, non-discrimination, the rights of the child, and consumer protection (Recital 81 DSA).

While the DSA requirements on systemic risks appropriately capture

some of the core challenges identified in the literature, several aspects seem left uncovered. For instance, the DSA and the interpretation of the General Court give little attention to systemic risks connected to the interaction between elements found within and external to a system, especially platforms as a system. Moreover, the background research and the literature scoped highlighted the role of data interconnectedness and concentration in the current market, clarifying explicit "data related practices" which could be connected to systemic risks under the DSA (Article 34(2)(e) DSA), as platforms and search engines have been explicitly recognised in their role of gathering, analysing but also monopolising data.¹⁵⁹

The literature helps in the contextualisation of the DSA recitals, and aids in finding examples of how technology and online services can cause an impact on the exercise of fundamental rights. Certain aspects of the themes and sub-themes in the literature may be however incomplete: The literature scoped has mainly brought clarity on how the design and functioning of VLOPs and VLOSEs can give rise to systemic risks. Relatively little attention has been given to the role of users and of their characteristics in the creation and spreading of systemic risks.

Both the literature and the DSA text nevertheless make clear how comparatively less quantitative the assessment of systemic risks is in the DSAs than in the financial sector. Except for the active user threshold discussed in Recital 76, the DSA embraces a qualitative assessment of systemic risks, even if it mandates VLOPs and VLOSEs to identify, measure, and mitigate risks (Article 34 DSA). The "measure" aspect is underdeveloped: there is no uniform framework to operationalise the systemic risk framework. Most of the literature scoped discusses systemic risks on the basis of the legal definition or on structural or contextual examples. Only one of the records in review proposes a framework to operationalise the measurement of systemic risks to fundamental rights. Moving beyond the binary "rights-based" approach, where a rights is either violated or not, and a reductionist "harm-based approach" that often limits itself to economic damages, Malgieri and Santos argue that interferences with fundamental rights should be viewed as a measurable spectrum ranging from mere social contact to severe violations.¹⁶⁰ To quantify the severity of these interferences, essential for compliance with the DSA, they introduce a three-tiered assessment methodology composed of objective, subjective and consequence-based parameters.¹⁶¹ For the first tier, they focus on the "objective normative evaluation", which assesses severity based on positivists legal factors, looking at the infringement of rules and principles that safeguard fundamental rights.¹⁶² In the second tier, they incorporate the "social meaning" of the infringement, evaluating society through the perspective of society, group identity and individuals.¹⁶³ In the third tier, they propose to assess the "real life tangible consequences" of the infringement on the lives of individuals and groups, including as key parameters economic or time loss, scale and reach, well-being or loss of opportunity.¹⁶⁴

¹⁵⁹ e.g., Bibri (n 89) 1350.

¹⁶⁰ Malgieri and Santos (n 77) 2.

¹⁶¹ *ibid* 7.

¹⁶² *ibid* 9.

¹⁶³ *ibid* 12.

¹⁶⁴ *ibid* 14–15.

¹⁵⁶ Global Network Initiative and Digital Trust and Safety Partnership (n 17).

¹⁵⁷ Peukert (n 13) 3; Leerssen (n 9); Buri and van Hoboken (n 8) 32; De Gregorio and Dunn (n 5) 487.

¹⁵⁸ *Amazon EU v Commission* (n 69) para 70.

The reason why the “measurement” aspect has been limited so far could be rooted in the acknowledging the *complexity* of measuring these systemic risks to fundamental rights, but also in the difference in terminology that has been adopted in literature so far. While for instance “systemic risk” is a very common concept in the financial frameworks, researchers in the field of privacy, social media, or disinformation have discussed “chilling effects,” “cascades,” or “virality.”¹⁶⁵ The novelty in the use of the term “systemic risks” in the context of online platforms and search engines has foreseeably impacted the amount and quality of literature scoped within the literature search.

5.2. Developing the qualitative framework

We highlighted several gaps in both the text of the DSA and in the academic literature discussing systemic risks to fundamental rights in the digital environment. These gaps not only reflect the conceptual novelty and interpretative ambiguity of systemic risks under the DSA, but also the lack of a structured method to identify, assess, and operationalise these risks. Tying this back to our original research question (“When does a risk become systemic?”), we developed a framework addressing these challenges. This framework directly assists a qualitative assessment of the “systemic” dimension of risks, grounded in the preamble and legal obligations of the DSA and informed by the thematic analysis conducted through our scoping review. This ultimately aids in identifying whether a determinate risk is systemic.

Established methodological approaches in related fields have informed the development of our framework, which we discuss here to complement the limited results of the scoping review. Within the field of data protection, Van Dijk et al. have critically examined the operationalisation of Data Protection Impact Assessments (DPIAs) and the relationship between risks and fundamental rights.¹⁶⁶ They view the approach to DPIAs framed as a risk management exercise as reductive and call for a “more socially robust assessment of technological impacts” that goes beyond mere risk calculation and includes diverse forms of knowledge, including insights of the larger public or data subjects, but also legal, scientific and technical expertise.¹⁶⁷ The value of an assessment of risks to fundamental rights that is more comprehensive compared to traditional risk assessment methodologies is also recognised by Yeung and Bygrave: Fundamental rights are “abstract social constructs” which safeguard “intangible moral values” rather than being primarily concerned with quantifiable or tangible harms.¹⁶⁸ Within the field of artificial intelligence, several frameworks have been recently introduced to assess the challenges posed by AI systems. Cosentini et al. develop a structured and comprehensive framework to tackle Fundamental Rights Impact Assessments (FRIAs) under the AI Act.¹⁶⁹ Through a qualitative questionnaire and semi-quantitative matrix, they aim to

gather both “contextual and operational insights” into high-risk AI systems, producing a risk indicator that then interacts with the “severity” and “probability of occurrence” impact score on all rights enshrined in the EU Charter of Fundamental Rights.¹⁷⁰ Mantelero proposes a methodological framework for the implementation of FRIAs within the AI Act which stresses the importance of a contextual, expert-based approach and that incorporates risk quantification in impact assessments, using a risk index for each potentially impacted right based on a likelihood and severity matrix.¹⁷¹ Novelli et al. contribute to the development of risk assessment methodologies by proposing a semi-quantitative approach, consisting of a scenario-based risk assessment model to assess AI risk magnitudes and integrating insights from climate change risk assessment and from legal proportionality tests.¹⁷² Last, Human Rights Impact Assessment (HRIA) methodologies to assess potential and actual adverse impact of business activities on fundamental rights offer valuable insights in developing a framework to investigate systemic risks to fundamental rights.¹⁷³

5.3. Assessing the “systemic” threshold of risks through the qualitative framework

The framework introduces five consecutive steps to guide in the evaluation of whether a given risk is systemic in character and groups them into three thematic classes: risk identification, risk impact, and risk probability. Each step is structured around a set of lead questions and example factors for consideration during the qualitative analysis. The questions provided under each phase of analysis are not exhaustive, but act as cues to help classify the risk and evaluate its breadth, depth, and complexity based on the text of the recitals and of the literature scoped. The more questions are answered in the affirmative, the stronger the indication that the risk under review is systemic. The steps and lead questions are conceptualised in Table 5.

Risk identification. Once a risk has been pinpointed, the risk identification class serves to locate the origin of this risk and anchor it within the normative framework of fundamental rights protection of the DSA (Article 34(1)(b) DSA). As a prerequisite, the risk should be connected to the “design, functioning and use” of designated VLOPs or VLOSEs and be relevant for the European Union context and society. After this, to contextualise the risk, the contributing sources of systemic risks should first be identified (Step 1: determine the origin). Four important domains to assess are: 1) data quality and availability, 2) data concentration, 3) design and deployment of the algorithmic systems, and 4) user behaviour. If the risk origin is connected to multiple areas, or if the origin remains ambiguous, this may be evidence of systemic risks and indicate integration and co-dependence between features. Second, the identified risk should be linked to specific protected rights and interests, as protected in the EU legal system through the Charter of Fundamental Rights (Step 2: determine rights impacted).¹⁷⁴ Article 34 DSA presents a

¹⁶⁵ see, e.g., Jonathon Penney, “Understanding Chilling Effects” (2022) 106 Minnesota Law Review 1451; Soroush Vosoughi, Deb Roy and Sinan Aral, “The Spread of True and False News Online” (2018) 359 Science 1146.

¹⁶⁶ Niels van Dijk, Raphaël Gellert and Kjetil Rommetveit, “A Risk to a Right? Beyond Data Protection Risk Assessments” (2016) 32 Computer Law & Security Review 286.

¹⁶⁷ *ibid* 296.

¹⁶⁸ Karen Yeung and Lee A Bygrave, “Demystifying the Modernized European Data Protection Regime: Cross-Disciplinary Insights from Legal and Regulatory Governance Scholarship” (2022) 16 Regulation & Governance 137, 143.

¹⁶⁹ Andrea Cosentini and others, “Assessing the Impact of Artificial Intelligence Systems on Fundamental Rights” [2025] MediaLaws <<https://www.medialaws.eu/assessing-the-impact-of-artificial-intelligence-systems-on-fundamental-rights/>> accessed June 1, 2025; Regulation (EU) 2024/1689 of the European Parliament and of the Council of 13 June 2024 laying down harmonised rules on artificial intelligence and amending Regulations (EC) No 300/2008, (EU) No 167/2013, (EU) No 168/2013, (EU) 2018/858, (EU) 2018/1139 and (EU) 2019/2144 and Directives 2014/90/EU, (EU) 2016/797 and (EU) 2020/1828 [2024] OJ L 2024/1689.

¹⁷⁰ Cosentini and others (n 169) 24–25, 35; Charter of Fundamental Rights of the European Union [2012] OJ C 326 (CFR) 2012.

¹⁷¹ Alessandro Mantelero, “The Fundamental Rights Impact Assessment (FRIA) in the AI Act: Roots, Legal Obligations and Key Elements for a Model Template” (2024) 54 Computer Law & Security Review 106020, 15.

¹⁷² Claudio Novelli and others, “AI Risk Assessment: A Scenario-Based, Proportional Methodology for the AI Act” (2024) 3 Digital Society 13.

¹⁷³ see, e.g., The Danish Institute for Human Rights, “Human Rights Impact Assessment; Guidance and Toolbok” (2020) 10 <https://www.humanrights.dk/files/media/dokumenter/udgivelser/hria_toolbox_2020/eng/dihr_hria_guidance_and_toolbox_2020_eng.pdf> accessed June 1, 2025; Désirée Abrahams and Yann Wyss, “Guide to Human Rights Impact Assessment and Management (HRIAM)” (International Business Leaders Forum (IBLF); International Finance Corporation (IFC) 2010) 18 <https://d306pr3pise04h.cloudfront.net/docs/issues_doc%2Fhuman_rights%2FGuidetoHRIAM.pdf> accessed June 1, 2025.

¹⁷⁴ see Article 34(1)(b), Recital 81 DSA and Charter of Fundamental Rights of the European Union [2012] OJ C 326 (CFR).

list of rights that can be used as a starting point, including human dignity, respect for private and family life, protection of personal data, freedom of expression and information, non-discrimination, rights of the child, right to a high-level of consumer protection.

Risk impact. Having reflected on where the risk originates and which rights it might affect, the risk impact class shifts focus to analysing the scale and remediability of the impact, in line with Recital 79 DSA. The scale of impact is assessed across three dimensions: individuals, society, and the economy (Step 3: determine scale of impact). If a risk impacts more domains, and the effects are diffused or compounding, this is likely to be systemic. To consider remediability, one should focus on the durability and reversibility of the harm (Step 4: determine remediability of impact). Important aspects to take into account include whether addressing the risk would require fundamental changes to how the VLOP or VLOSE operates, whether the risk is self-perpetuating or whether legal or technical remedies are available, and whether its consequences might have a long-lasting impact on fundamental rights.

Risk probability. The risk probability class turns to assessing the likelihood that the identified risk will materialise or persist over time (Step 5: determine the probability). Systemic risks are often connected to features that are inherent to VLOPs or VLOSEs operations or user practices, which are more likely to persist and scale over time. It should be considered whether the system or practice is integrated into society, whether regulatory safeguards are available, whether there are known patterns of risk, or financial incentives for the risk.

While progressing through the phases, the reviewer builds a cumulative profile of the risk. By considering how each aspect of the risk adds to its nature, the reviewer gains, at the end of the assessment, a clearer understanding of the characteristics of the risk at stake: what it is, where it originated, who and what it affects, whether it can be mitigated, and how likely it is to occur. If a combination of all elements suggests a risk that stems from multiple or ambiguous origins, that affects fundamental rights, that produces broad and cascading harm extending beyond individuals to societal structures or economic systems, that is difficult to remediate without structural change, and that is likely to persist over time, the reviewer can, through answering the questions of the framework understand that the risk meets the threshold of “systemic.”

6. Conclusions

This article analysed when a risk becomes “systemic” according to existing academic literature. Although systemic risks are a key part of the DSA risk assessment framework, the Regulation does not provide a

clear definition or guidance to make this notion operational. This ambiguity is particularly challenging as systemic risks have been primarily dealt with in the financial sector, and only with the DSA has this discourse extended to online platforms and search engines.

The “systemic” threshold has been examined through a scoping review and thematic analysis of literature published until November 2025 discussing fundamental rights risks to systems. We have seen how in digital environments systemic risks arise from the integration of technology into societal structures, which are now so interconnected that they evolve and shape one another.¹⁷⁵ Accordingly, we generated three themes to make the “systemic” threshold explicit: we identify structural characteristics of systemic risks, sources of systemic risks – including quality and availability of data, data concentration, the design and use of algorithms, and the behaviour of users of technologies –, and last, impact of systemic risks– on individuals, society, or the economy. Building upon these findings and the legal requirements to establish “systemic risks” under the DSA we developed a qualitative framework to help researchers, policymakers and platforms assess systemic risks, constituting a structured basis for determining if a risk meets the threshold of systemic.

The framework should be understood as a non-exhaustive analytical tool, intended to guide the reflection on whether a particular identified risk is systemic. Future refinement would benefit from engagement with industry stakeholders and civil society to ensure that the framework can be integrated with quantitative consideration to help also objectively identify, assess and measure the risk and its effect, and that it captures the practical challenges that VLOPs and VLOSEs currently face in managing systemic risks.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Annex I

¹⁷⁵ Strauß (n 70) 18.

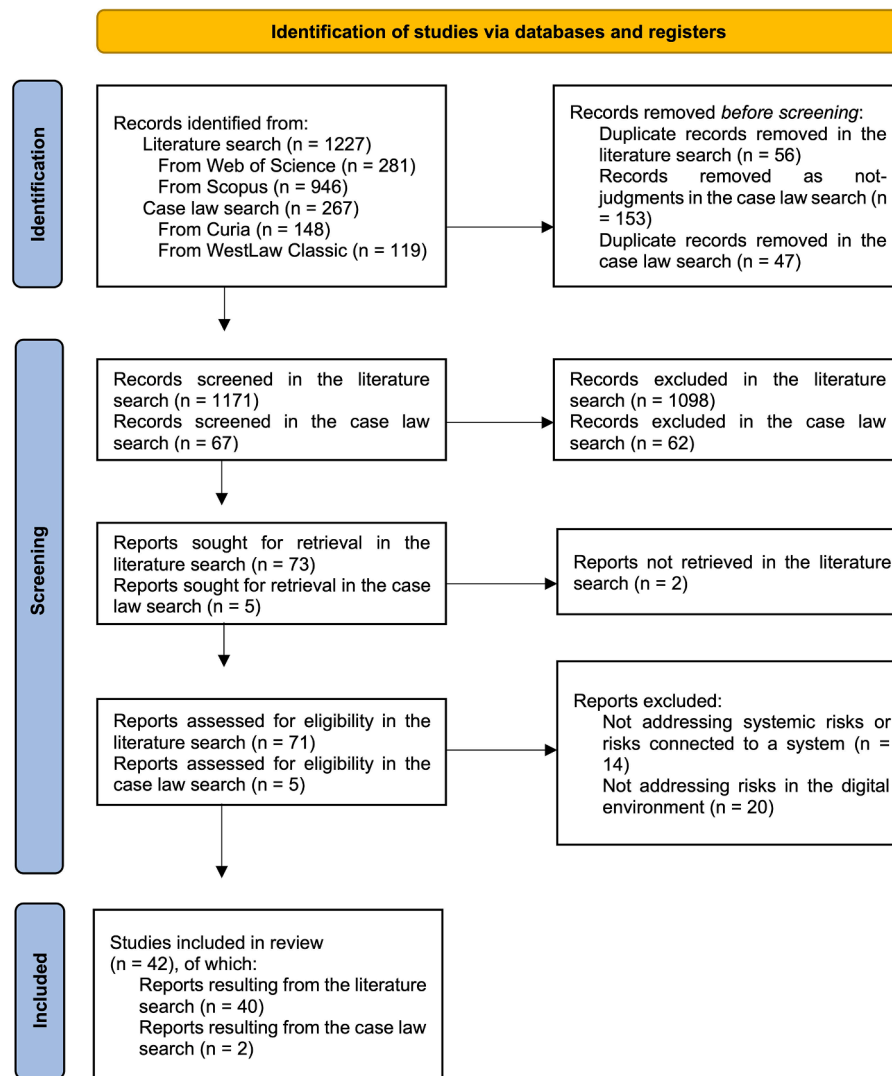


Fig. 1. PRISMA Flow Diagram representing the evolution of the body of literature scoped during the different stages of the scoping review. It shows the number of records that were first collected, then screened, and finally included in the review. It also explains why records were progressively excluded.

Annex II

Table 6

Table of studies included in the review, organised per author, year, title, publication and type of publication.

Author	Year	Title	Publication	Type of publication
Arifin et al.	2025	Designing an AI-Driven Legislation Framework to Improve Indonesia's Law-Making Transparency and Public Participation	Yustisia Jurnal Hukum	Article
Aseeva & Budanov	2020	Digitalization: Potential risks for civil society	Economic Annals-XXI	Article
Beltrán	2025	AI algorithms under scrutiny: GDPR, DSA, AI Act and CRA as pillars for algorithmic security and privacy in the European Union	Computers & Security	Article
Berridge & Grigorovich	2022	Algorithmic harms and digital ageism in the use of surveillance technologies in nursing homes	Frontiers in Sociology	Article
Bibri	2023	The Metaverse as a Virtual Model of Platform Urbanism: Its Converging AIoT, XReality, Neurotech, and Nanobiotech and Their Applications, Challenges, and Risks	Smart Cities	Article
Bishop	2021	Influencer Management Tools: Algorithmic Cultures, Brand Safety, and Bias	Social Media and Society	Article
Chan	2025	Reviewing and reconceptualising cyberbullying through a social and sustainable lens	The Social Science Journal	Article
Chowdhury	2024	Technology is never neutral: Robodebt and a human rights analysis of automated decision-making on welfare recipients	Australian Journal of Human Rights	Article
Correia de Carvalho	2025	It will be what we want it to be: Sociotechnical and Contested Systemic Risk at the Core of the EU's Regulation of Platforms' AI Systems	Journal of Intellectual Property, Information Technology and E-Commerce Law	Article

(continued on next page)

Table 6 (continued)

Author	Year	Title	Publication	Type of publication
Crocker	2020	The Fourth Amendment at Home	Indiana Law Journal	Article
Currie et al.	2025	Rethinking technology regulation in the age of AI risks	Journal of Information Technology	Article
Eder	2024	Making Systemic Risk Assessments Work: How the DSA Creates a Virtuous Loop to Address the Societal Harms of Content Moderation	German Law Journal	Article
Ferguson	2017	Big data surveillance: The convergence of big data and law enforcement	The Cambridge Handbook of Surveillance Law	Book chapter
Gopalan et al.	2023	Financial globalisation in ASEAN+3: Navigating the financial trilemma	Global Policy	Article
Grant & Smith	2021	Activism in the time of COVID-19	Group Processes and Intergroup Relations	Article
Haitsma	2025	Addressing discrimination in algorithmic profiling: Examining risk governance in Dutch public social security agencies	European Journal of Social Security	Article
Helm & Seubert	2020	Normative Paradoxes of Privacy: Literacy and Choice in Platform Societies	Surveillance & Society	Article
Hill	2023	Navigating visibility and risk: disabled young women's self-presentation practices on social media	Journal of Gender Studies	Article
Jang	2025	Peeling Back the Layers of "Paint on Rotten Wood": Unraveling the Senate's "Big Tech and Child Sexual Exploitation Crisis" Hearing	International Journal of Communication	Article
Kaesling & Wolf	2025	Sustainability and Risk Management under the Digital Services Act: A Touchstone for the Interpretation of 'Systemic Risks'	GRUR International	Article
Kaviani Johnson & Sloth-Nielsen	2020	Child protection, safeguarding and the role of the African charter on the rights and welfare of the child: Looking back and looking ahead	African Human Rights Law Journal	Article
Kusche	2024	Possible harms of artificial intelligence and the EU AI act: fundamental rights and risk	Journal of Risk Research	Article
Lee	2020	Access to Finance for Artificial Intelligence Regulation in the Financial Services Industry	European Business Organization Law Review	Article
Lendvai	2024	Taming the Titans? – Digital Constitutionalism and the Digital Services Act	ESSACHESS - Journal for Communication Studies	Article
Lucchi	2013	Freedom of expression and the right of access to the Internet: A new fundamental right?	Routledge Handbook of Media Law	Book chapter
Malgieri & Santos	2025	Assessing the (severity of) impacts on fundamental rights	Computer Law & Security Review	Article
Marbin et al.	2021	Avoiding the Virtual Pitfall: Identifying and Mitigating Biases in Graduate Medical Education Videoconference Interviews	Academic Medicine	Article
McAuliffe et al.	2021	Digitalization and artificial intelligence in migration and mobility: Transnational implications of the covid-19 pandemic	Societies	Article
McMahon et al.	2020	Big Data Governance Needs More Collective Responsibility: The Role of Harm Mitigation in the Governance of Data Use in Medicine and Beyond	Medical Law Review	Article
Ó Fathaigh et al.	2025	The Regulation of Disinformation Under the Digital Services Act	Media and Communication	Article
Palumbo	2024	A Medley of Public and Private Power in DSA Content Moderation for Harmful but Legal Content: An Account of Transparency, Accountability and Redress Challenges	Journal of Intellectual Property, Information Technology and E-Commerce Law	Article
Papathanasopoulou	2025	From professional fact-checkers to the crowd: can Meta's Community Notes survive the Digital Services Act (DSA)?	Information & Communications Technology Law	Article
Pavone et al.	2016	A systemic approach to security: Beyond the tradeoff between security and liberty	Democracy and Security	Article
Quigley & Roy	2012	Cyber-Security and Risk Management in an Interoperable World: An Examination of Governmental Action in North America	Social Science Computer Review	Article
Röösli et al.	2022	Peeking into a black box, the fairness and generalizability of a MIMIC-III benchmarking model	Scientific Data	Article
Sekwenz & Gsenger	2025	The Digital Services Act: Online Risks, Transparency and Data Access	Digital Decade: How the EU Shapes Digitalisation Research	Book chapter
Strauß	2019	Privacy and Identity in a Networked Society: Refining Privacy Impact Assessment		Book
Xenidis	2020	Tuning EU equality law to algorithmic discrimination: Three pathways to resilience	Maastricht Journal of European and Comparative Law	Article
Zetsche et al.	2020	The Evolution and Future of Data-Driven Finance in the EU	Common Market Law Review	Article
Zhu & Song	2022	Systematic Regulation of Personal Information Rights in the Era of Big Data	SAGE Open	Article

Table 7

Table of quotations from the studies analysed through the Braun & Clarke (2006) thematic analysis, organised per theme and subtheme.

Theme & sub-topic	Authors/Case	Interesting statements/arguments of the respective authors
Structural characteristics of systemic risks	Correia de Carvalho, 2025	"The risk-based approach to AI regulation adopted in the DSA and AIA will face two main challenges. The first is conceptual: risk is often conceived in an actuarial and individual fashion, i.e., it focuses on quantitatively identifying and assessing risks of harm caused to specific individuals or entities. These dominant conceptions of risk, while easier to calculate, fail to fully capture less quantifiable and intangible AI risks – e.g., to democracy, fundamental rights, civic discourse, or of gender-based violence – whose perceptions are contestable and highly subjective but that both the DSA and AIA aim to address." (p. 37)
	Palumbo, 2024	"As it determines the scope of the limitation on freedom of expression, the interplay between public influence and private ordering has important consequences for the freedom of expression of millions of recipients of VLOPs and VLOSEs' services. This interplay creates a new dimension of power and interference which may be difficult to categorise under traditional constitutional concepts." (p. 256)
Sources of systemic risk: Quality and availability of data	Bibri, 2023	"The phenomenon of platformization has recently risen to prominence and given rise to what has been termed a "platform urban society", a digitally connected urban world where platforms have penetrated virtually every

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Table 7 (continued)

Theme & sub-topic	Authors/Case	Interesting statements/arguments of the respective authors
Sources of systemic risk: Data concentration		aspect of urban life. As a result, the data infrastructures, governmental frameworks, and institutional processes of digital platforms are rapidly transforming the different sectors and everyday-life domains of urban societies, as well as reorganizing practices, reconstructing imaginaries, and reshaping discourses around these platforms" (p. 1350)
	Ferguson, 2017	"In daily practice, private big data companies are suctioning up vast streams of consumer data to target individuals or families for commercial gain. Law enforcement agencies are building information centers to collect, aggregate, and disseminate criminal records data and other forms of biometric data and locational data for investigative advantage. While developing along separate evolutionary paths, these collection and aggregation systems have begun overlapping in practice. Individuals share personal data with private companies. Private companies sell personal data to law enforcement. Law enforcement integrates this information with publicly available data. Databases merge, blend, and share related private-public data." (p. 171)
	Gopalan et al., 2023	"Since Big Techs hold vast amounts of consumer data in their servers, data sovereignty and misuse of local consumer data is also a big problem for those countries which currently do not have a consumer data protection regime in place." (p. 473)
	McMahon et al.	"The 'pervasiveness' of data means that in the context of large, digital, connected or connectable, and often fast-changing datasets, harms are often systemic and cannot be captured by linear cause and effect." (p. 156)
	Strauß, 2019	"Today we may speak of a networked society as networking structures occur within and between offline and online environments in many respects, enabled and reinforced by ICTs. Hence, these shifts did not merely affect how information is structured and processed in digital environments. Society has increasingly entered a stage of convergence between analog and digital environments with information as a driving force in our whole ecosystem" (p. 4)
	Zetzsche et al., 2020	"Where the power is in the data, we would recommend that financial regulators focus on the new systemic and other risks stemming from concentration of data in the hands of a few technology firms, as a complement to traditional approaches to systemic risk represented by banks that were too-big-too-fail or too-connected-too-fail." (p. 355)
	Zhu & Song, 2022	"The boom in mobile internet and big data technology has produced significant changes to the means and the extent of personal information collected. While enjoying the conveniences of user-customized services and targeted marketing that big data affords, people also have seriously suffered from personal information leakage and a seemingly unavoidable infringement upon their privacy." (p. 1)
	Grant & Smith, 2021	"Given that computer-mediated communication can make social categories more and interpersonal differences less salient, the emotional content of the memes and posts that people tend to share is likely to intensify group-based emotions. Hence any negative, group-based emotional reaction to an injustice, such as an incidence of police brutality, is more likely to be amplified by the prevailing emotional climate and sweep through the affected community." (p. 301)
	Helm & Seubert, 2020	"Algorithmic processing of data opens up endless possibilities to search for patterns and to create new combinations, correlations, and profiles. This offers enormous opportunities for optimization and profit but also challenges for the protection of privacy both in regards to the state and the private sector. In the last ten years, huge storage and evaluation capacities have been developed with which the control capacities of the democratic constitutional states of the world cannot keep pace, partly because future uses of data can hardly be foreseen" (p. 188)
	Lucchi, 2013	"The relevance of networked communication as a tool of mass democracy is increasingly evident. In some countries, the Internet is the one of very few sources of pluralistic and independent information. The events of the Arab Spring have served to highlight how important new communication and information technologies have become. Using a mix of blogs and social networking sites, the new medium has demonstrated its power to support spontaneous democratic mobilization from below: a concrete and participatory form of democracy. The result of these online movements was surprising, with hundreds of thousands of people being summoned to action. Up to now this kind of influence was a prerogative that belonged only to political and union organizations. The impact that digital communication tools can have on public opinion and decision-making is therefore enormous" (p. 158)
Sources of systemic risk: Design and use of algorithms	Berridge et al., 2022	"Surveillance technologies thus pose a disproportionate risk to residents' self-determination by characterizing them as threatening to others and by extending the power of care workers and administration to control their live" (p. 7)
	Bibri, 2023	"There are various processes that enable the practice and functioning of platformization. The key one is algorithmization, a process by which most urban activities and processes are algorithmized by converting their informal description into a set of computer-programmed step-by-step instructions that can be used to perform a large-scale computation using mathematical and logical models for calculating specific objective functions. (...) These include, for example, predicting and modifying behaviors, inferring social or political affiliations, manipulating thoughts, influencing attitudes, affecting mental and emotional states, modulating neurological activities, editing genomes, and making decisions on behalf of users." (p. 1351)
	Bishop, 2021	"What the industry calls marketing 'solutions,' hereafter referred to as influencer management tools, present analytics data and make algorithmic calculations designed to support marketers in selecting appropriate influencers for advertising campaigns, through subjective calculations about influencers' brand safety and risk." (p. 1)
	McAuliffe et al., 2021	"While the challenges posed by the development of digital technologies and AI are not exclusive to migration, the power imbalance between migrants and public authorities may aggravate existing weaknesses in protecting their human rights. Two areas are of particular interest—the potential for algorithmic bias to exacerbate inequalities and lead to unlawful discrimination and the effects of increased surveillance on the protection of migrants' right to privacy." (p. 8)
Sources of systemic risk: Users of technology	Johnson & Sloth-Nielsen, 2020	"This also has implications for child safeguarding. In 2020 the African Children's Committee is debating a draft General Comment on article 27 of the African Children's Charter, which elaborates measures required to be taken by state parties to combat 'offline' and online sexual exploitation and abuse of children. As explained in the African Children's Committee's General Comment 5, child protection systems that contain the full array of measures to prevent and respond to all forms of child abuse and exploitation are integral to the implementation of the African Children's Charter, including to address forms of violence not contemplated at its inception." (p. 664)

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Table 7 (continued)

Theme & sub-topic	Authors/Case	Interesting statements/arguments of the respective authors
Societal impact of systemic risks: Society	Marbin et al., 2021	“While video conference interviews (VCIs) can be both convenient and cost effective, they may also exacerbate existing biases and introduce new sources of bias into the recruitment process. Graduate medical education (GME) programs committed to equity and diversity in recruitment must be aware of these biases and implement strategies to mitigate them. ... “Digital redlining,” or decreased access to broadband internet in marginalized communities, is linked to existing structural racial and/or socioeconomic disparities. Digital redlining may limit access to broadband internet for some applicants. In addition, applicants may not be able to afford broadband connections or high-quality cameras, microphones, or computers.” (p. 1120–1121)
	Aseeva & Budanov, 2020	“The process of transition to the digital economy has been ongoing for the past ten years amid the financial and economic crisis of the world civilization involving restructuring of industries, transport and agriculture, and giving new opportunities for work with Big Data and optimization of processes in the economy due to decision-making, developing smart environments, cryptocurrencies and hybrid forms of interaction between human beings and artificial intelligence.” (p. 38)
	McAuliffe et al., 2021	“Substantial developments in computational power and data generation have resulted in people’s daily lives being shaped through AI-related services and digital interfaces, such as the personalization of newsfeeds, online advertising tailored to browsing activity, interactive chatbot assistance and automated analytical updates.” (p. 1)
	McMahon et al., 2020	“Digital data are also multiple in the sense that they can be in more than one place at once, and that they can be interlinked in various ways, which makes it extremely difficult to trace the movements of data in specific cases.” (p. 157)
	Lucchi, 2013	“The rapidly evolving media revolution has generated a number of new regulatory initiatives designed to reduce systemic risks associated with this means of communication, “ranging from risks to children, to privacy, to intellectual property rights, to national security, which might more indirectly, and often unintentionally, enhance or curtail freedom of expression” (p. 157)
	Pavone, 2016	“Given its almost unrestrained and unaccountable deployment, it has become a significant threat for the freedom of expression and for a free political debate, and therefore a threat for democracy and, ultimately, for the society itself. In fact, it has reduced the overall security levels enjoyed by Western societies and has undermined citizens’ trust in both political and legal institutions.” (p. 15)
	Quigley & Roy, 2012	“Widening degrees of interoperability thus open up opportunities for innovation, democratization, and “mass collaboration”. Yet the same time, whether or not the cyber-infrastructure that results is secure is perhaps the key determinant for the future of e-government, e-commerce, and digital life more generally.” (p. 84)
	Strauß, 2019	“Systems theory provides a methodological approach to cope with this complexity through different layers of abstraction and thus allows the bigger picture of sociotechnical change to be grasped. On the one hand, conceptualizing a phenomenon as a system allows focusing on the interplay of its interrelated components including their dynamics. On the other hand, it also supports investigation of how a system is related to external factors in its environment (e.g., how a technology pervades and transforms specific domains of society). This supports the analysis of transformation patterns of socio-technical phenomena and the assessment of societal impacts resulting from them, such as those represented by ICTs.” (p. 17)
	Zetsche et al., 2020	“At the same time, rapid evolution in FinTech is raising new risks. The sheer amount of data facilitates looking at correlations rather than causations, and correlations can lead to unintended, and socially regressive, consequences. For instance, algorithms can discriminate wrongfully against certain groups of people: if data analytics show a certain race or gender generally has a better credit score, that better score could derive from existing biases against other races or genders, or the algorithm could be programmed to analyse data in a certain way that reflects biases existing among developer groups, or their clients, respectively.” (p. 335)
	Zhu & Song, 2022	“The development of technology has always been an important factor giving rise to the change of legal theory and system. Legal theory and system should pay attention to and respond to new problems brought by technology, and must not hang on to what’s outmoded.” (p. 3)
	Berridge & Grigorovich, 2022	“The expansion of surveillance technologies for contact tracing in nursing homes also has implications for the ways care workers are directed, evaluated, and disciplined. As with residents, interest in these technologies for monitoring workers in nursing homes pre-dates the pandemic and is part of a broader phenomenon of algorithmic management across sectors such as retail, manufacturing, banking, hotels, and call centers” (p. 8)
	Bibri, 2023	“And what this mode of governance entails in terms of social inequality, discrimination, mass control, unfairness, technocratic reductionism, governmentality, and democratic decay embedded in algorithms, recommendation systems, and engagement strategies.” (p. 1367)
Societal impact of systemic risks: Individuals	Bishop, 2021	“Long social histories of discriminatory decision-making are baked into engineering practices, training datasets, and underscore patterns in predictive algorithmic modeling.” (p. 5)
	Ferguson, 2017	“This personal data is regularly bought, sold, and repurposed by a whole network of advertisers, ad exchanges, and market researchers. As Thomas Hemmes has described: One might think of a person’s digital identity by analogy to a pointillist painting. Thousands upon thousands of tiny bits of digital information about an individual, including what we have called basic facts, sensitive facts, and transactional facts, can be assembled to form a picture of the individual: his likes, dislikes, predispositions, resources; and in fact, any facet of his personality that has had contact with the Internet.” (173–174)
	McAuliffe et al., 2021	“Bias may arise from a lack of representation and diversity in datasets used to train AI algorithms. For example, researchers found that due to a lack of diversity in the datasets used for training commercially available facial recognition algorithms, these were significantly less accurate when recognizing women’s faces with darker skin types.” (p. 8)
	McMahon et al., 2020	“Predictive analytics could also lead to poor consumer ratings which in turn result in the denial of mortgages; predictive health profiles could make finding employment more difficult or raise insurance premiums, etc. Studies have also shown the risks of ‘credit worthiness by association’ such as where an individual’s credit card limit was reduced due to predictions based on repayment histories of other people who shopped in the same stores as that individual had” (p. 161)
	Röösli et al., 2020	“The expansive availability of digital health data has led to a colossus of data-driven models to guide and improve healthcare delivery. This change of paradigm will and partially already does decisively shape medical diagnostics, drug discovery, clinical research and personalized medicine. These tools approach and sometimes surpass expert clinicians in certain specialties. Thereby, medical artificial intelligence (AI) has the potential to render healthcare more efficient and effective through better informed decisions and improved patient outcomes. However, emerging evidence suggests that many of these data-driven clinical decision support tools

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Theme & sub-topic	Authors/Case	Interesting statements/arguments of the respective authors
Societal impact of systemic risks: Economy		may be biased and not equally benefit all populations. As AI makes progress to improve quality of care for some patients, others are left behind. Particularly minorities and historically disadvantaged groups are at risk of suffering from unfair model predictions, as we have seen for example in the case of COVID-19" (p. 1)
	Strauß, 2019	"Given the peculiarities of digital information, it can create a sort of permanent link to the identity of an individual person. This has effects on the privacy of this person as well: "one's informational sphere and one's personal identity are co-referential, or two sides of the same coin" (p. 6)
	Xenidis, 2020	"A first fundamental incompatibility between algorithmic discrimination and the grammar of nondiscrimination law relates to the problem of intersectionality. Inequalities are complex and multidimensional and so are phenomena of discrimination, a reality which is reflected in the data that feeds algorithms. This social complexity clashes with the organization of non-discrimination law around single distinct protected grounds, for instance gender, race, age, disability, sexual orientation or religious beliefs. Critical legal theory scholars have widely recognized the inadequacy between the prevailing 'single-axis' non-discrimination law model and the complexity and compounded nature of inequalities and discrimination" (p. 739)
	Zhu & Song, 2022	"The boom in mobile internet and big data technology has produced significant changes to the means and the extent of personal information collected. While enjoying the conveniences of user-customized services and targeted marketing that big data affords, people also have seriously suffered from personal information leakage and a seemingly unavoidable infringement upon their privacy." (p. 1)
	Gopalan et al., 2023	"Greater data sharing among firms and with regulators can alleviate financial instability by increasing transparency and enabling policymakers to calibrate financial policies in an informed manner. For example, credit registries maintained by regulators reduce information asymmetries and enable firms to extend credit to borrowers depending on their creditworthiness." (p. 473)
	Lee, 2020	"Consumers are usually deemed to be vulnerable, as they do not understand the mechanisms working in financial markets and also because they cannot fully understand the advice provided." (p. 742)
	Zetzsche et al., 2020	"We support market structure-related interventions which aim to maintain the independence of, and choice among, critical infrastructure providers as well as data portability rights in favour of financial customers, in order to comply with overarching objectives such as financial stability, financial innovation, development and inclusion, financial integrity, and consumer protection." (p. 355–356)
	Gopalan et al., 2023	"These concerns are exacerbated in case of algorithmic and AI/ML techniques deployed by fintechs and Big Techs to make decisions about the provision of financial services to different sections of the population. Whilst these approaches may contribute to preserving financial stability by mechanising lending decisions, they could also systemically undermine consumer welfare by systematically excluding a particular section of the society, thereby carrying implications for financial inclusion as well." (p. 473)

Data availability

Data will be made available on request.